



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

PHYSICAL AND CHEMICAL CHANGES

ATOM:

It is simplest form of matter and therefore cannot be split into simple substances by any chemical (or) physical method.

(or) An indivisible particle

E.g. Hydrogen - H - The lightest gas / element

Helium - He - The lightest Nobel gas

* Rubidium - Rb - it is used to make atomic clocks

* Cesium - Cs - it is used to make atomic clocks

* Lithium - Li - The lightest metal

* Strontium - Sr - The responsible element for Red fire work

Calcium - Ca - The responsible element for Orange fire work

Sodium - Na - The responsible element for Yellow fire work

Barium - Ba - The responsible element for vivid green fire work

MOLECULE:

Group of atoms are called molecule.

(1) $H+H \rightarrow H_2$ (cryogenic fuel - Liquid H_2 + Liquid O_2)

(2) $O+O \rightarrow O_2$ (O_2 is a paramagnet)

(3) $O_2+O \rightarrow O_3$ (O_3 is a Diamagnet)

SUBSTANCE (OR) MATTER:

Something that occupies space, possess mass and offer resistance to any stress applied on it.

(or) Group of molecules

Eg: (1) Silver (Ag) + Iodine (I_2) $\rightarrow$ AgI (Silver Iodide)

$\rightarrow$ The compound used in cloud seeding (or) Artificial Rain

(2) Sodium (Na) + Chlorine (Cl_2) $\rightarrow$ NaCl (sodium chloride)

IMPORTANCE OF SODIUM CHLORIDE

$\rightarrow$ Common names: Table salt, Common salt, Rock salt, Sea Salt

$\rightarrow$ Sea water contains about 2.9% NaCl.

$\rightarrow$ It is used as preservation of food articles like fish meat etc.

$\rightarrow$ used to make freezing mixture.

$\rightarrow$ impure salt (Rock salt) is used to de-Ice Roads (Removal of snow, ice or frost from a surface).

FREEZING MIXTURE $\rightarrow$ Generally used to achieve temperature less than $0^\circ C$.
i.e below the freezing point of water.

Eg. Ice water and salt ($-50^\circ C$); dry ice + alcohol ($-80^\circ C$)

ARTIFICIAL RAIN (OR) CLOUD SEEDING:

The foreign particles which are drenched over these clouds can be Dry Ice, silver Iodide salt powder etc. This process is known as cloud seeding. This stimulation is done by aero planes or rockets and then finally 'Rain on' Super cooled chemicals - Dry Ice (or) Iodine.

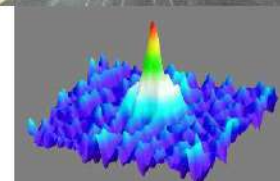
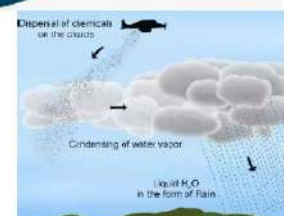
COMPOUND: (Homogeneous)

Compound is a pure substance made up of 2 (or) more elements combined in a definite proportion by mass.

E.g. $H_2 + 2 O_2 \rightarrow 2 H_2O$
 $1 N_2 + 3 H_2 \rightarrow 2 NH_3$

$\rightarrow$ Any two elements may combine in more than one way to give more than one compound.

Eg: $C + O_2 \rightarrow CO$
 $C + O_2 \rightarrow CO_2$



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON Google Play

Download on the App Store

MIXTURE: A mixture contains more than one substance.

Mixtures are constituted by more than one kind of pure form of matter, known as a substance.

Substance can't be separated into other kinds of matter by any physical process.

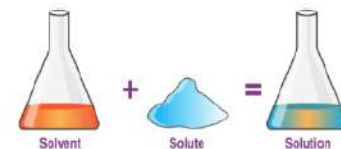
E.g.

Salt water → salt separated from H₂O by the Physical process of evaporation. However, NaCl is itself a substance and cannot be separated by physical process into its chemical constituents.

SOLUTION:

A solution is a Homogeneous mixture of two (or) more substances.

Solution = Solute (Lesser quantity of solution) + Solvent (Larger quantity of solution)



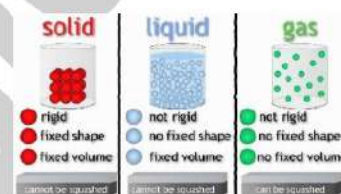
ALLOYS:

Alloys are mixtures of two or more metals or a metal and a non-metal and cannot be separated into their components by physical methods. But still, an alloy is considered as a mixture because it shows the properties of its constituents and can have variable compositions.

COLLOIDAL SOLUTIONS:

A heterogeneous mixture

E.g. Milk



PHYSICAL CLASSIFICATION OF MATTER:

The energy of this universe existed in 5 physical states.

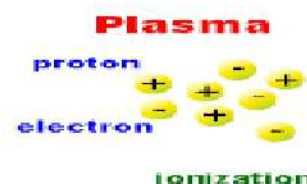
INTER MOLECULAR DISTANCE

- | | | | |
|-----|--------|--------------------------------|-------------|
| (1) | Solid | → Regular arrangement of atoms | - very low |
| (2) | Liquid | → Irregular arrangements | - High |
| (3) | Gas | → Irregular arrangements | - very high |

- (4) **Plasma** → Ionized form - existed at high temp & high pressure

E.g. Stars, Sun chromosphere, (6000°C), earth Core (6000°C)

Corona (6 lakhs °C), Neon sign Bulb, Fluorescent tube (He (or) some other gases)



(5) **BASE EINSTEIN CONDENSATE:**

In 1920, Indian Physicist Satyendra Nath Bose had some calculations for a fifth state of matter.

→ Predicted by (Bose-Einstein condensate) Einstein

→ The Nobel Prize in Physics 2001 was awarded jointly to Eric A. Cornell, Wolfgang Ketterle and Carl E. Wieman "for the achievement of Bose-Einstein condensation in dilute gases of alkali atoms, and for early fundamental studies of the properties of the condensates."

it is formed at a gas is cooling of Extremely low temperature i.e.

absolute zero (- 273.15 °C or '0' kelvin)

BASE - EINSTEIN THEORY:

HEAT: If the vibration of particles in a substance is High that state is considered as Hotness.

Eg:

Na - Cl

Na ~~~~~ Cl

vibrations - increases → Heat increases → Temperature ↑

Note: The velocity of gases is maximum at High temperature region only.

COLDNESS: If the vibration of particles is low that state can be considered as coolness.

Eg: Na - Cl

Na ~~~~ Cl

vibrations - decreases → Heat decreases → Temperature decreases



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON Google Play

Download on the App Store



BIOLOGICAL SPECIES CAN BE PRESERVED IN

- (1) Ice
- (2) Dry Ice (solid form of CO_2)
- (3) Formalin ($\text{HCHO} + \text{H}_2\text{O}$)
- (4) Mercury
- (5) Liquid N_2

Note:

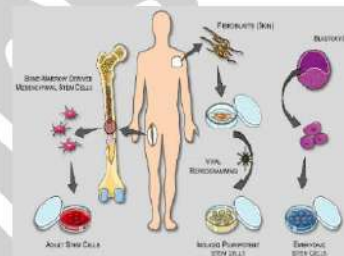
1) Around 4 lakh plant DNA samples are preserved by using cryogenic chemicals at NBPGR - National Bureau of plant Genetic Resources - Delhi

The standard temperature of seed bank for long term conservation of plant genetic resources - (-18°C)

2) **Stem cells:** it can be preserved in Liquid nitrogen (-196°C)

Potential use of stem cells:

- (1) Baldness (2) Blindness (3) Deafness (4) Diabetes
- (5) Multiple sites cancers (6) Myocardial infarction (7) Muscular dystrophy (8) Spinal cord injury (9) Bone marrow transplantation (currently established) (10) Wound healing (11) missing teeth (12) stroke (13) Traumatic brain injury (14) Learning defects (15) Alzheimer's disease (16) Parkinson's disease (17) Male infertility due to absence of spermatogonia stem cells.



CHEMICAL REACTION:

The change of chemical composition can be represented with symbols (and) formulas are termed as Chemical Reaction.



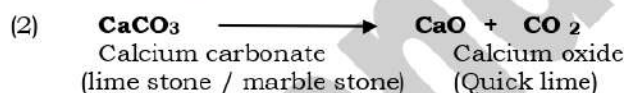
Ammonium Nitrate (an explosive compound)

Nitrous oxide (laughing gas)

Note: Some important anesthetic drugs are - N_2O , chloroform

Heat is a form of energy, when a substance is heated, the following changes can be observed.

Eg:



Base on above equations, following changes can be observed. Those are

- 1) Physical change
- 2) Chemical change

PHYSICAL CHANGE: (REVERSIBLE REACTION)

The inter conversion of states is a physical change because these changes occur without a change in composition and no change in the chemical nature of the substance.

(Or)

when a substance is Heated, the chemical composition cannot be changed. Those type of reactions are considered as physical change.

(or)

Changes occurred in colour, volume and physical state.

Note: Physical properties - Color, hardness, rigidity, fluidity, density, M.P., B.P., Physical state, volume





Chandan Logics

Chandan Logics

Aspire to Inspire

Download Chandan Logics APP

Contact: 96 76 57 8793

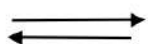
94 94 55 8793

GET IT ON

Google Play

Download on the

App Store

COLOUR:(1) Zinc Oxide (ZnO)
(white)Zinc oxide (ZnO)
(yellow)The common name of ZnO is **Philosopher's wool / flower of zinc**PbO
(yellow)PbO
(Brown)**PHYSICAL STATE:**

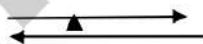
	Solid	Liquid	Gas	Plasma
Solid	-	Melting	Sublimation	-
Liquid	Freezing / Fusion	-	Boiling / Evaporation	-
Gas	Deposition	Condensation	-	Ionization
Plasma	-	-	Deionization	-

Deposition:Gas $\longrightarrow$ SolidE.g. conversion of water vapor to **frost** (subfreezing air)**Sublimation:**Solid $\longrightarrow$ Gas

- E.g.
- (1) Dry Ice (Boiling point -78.5°C)
 - (2) Iodine vapor (These are used for trace the finger prints)
 - (3) Naphthalene
 - (4) Camphor
 - (5) Menthol (cloves - Alkaloids)
 - (6) Benzoic acid / sodium Benzoate (used for food preservation)
 - (7) NH_4Cl - ammonium chloride (smelling salt)
 - (8) NH_4Br , NH_4I
 - (9) Arsenic at High temperature
 - (10) Sleeping pills (HgCl_2)

Note:

- 1) At High temperature $\text{NH}_4\text{Cl} \xrightarrow{\Delta} \text{NH}_3 + \text{HCl}$ (Chemical change)
- 2) wax $\xrightarrow{\Delta} \text{CO}_2 + \text{H}_2\text{O}$

VOLUME:Natural gas
Methane (CH_4)CNG (compressed natural gas)
methane (CH_4)**COMMON NAMES OF CH_4 (METHANE)**

- 1) Gobar gas
- 2) PNG (Piped Natural gas)
- 3) NG (natural gas)
- 4) Oil gas
- 5) LNG (Liquid Natural Gas)
- 6) Coal gas
- 7) Marsh gas
- 8) Biogas

Follow Chandan Logics on



Download Chandan Logics APP

Contact: 96 76 57 8793 / 94 94 55 8793

One STOP For ALL Competitive EXAMS

GET IT ON

Google Play

Download on the

App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

MARSH GAS : It is a poisonous gas & releases from Coal mines .

To find out the leakages of marsh gas the device used is Ansel's Alarm (Grahams law $r = 1/\sqrt{d}$)

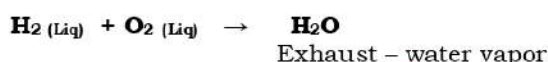
Note : Cooking gas (LPG) (or) Lighters = Butane + Iso butane (Propylene - 2.5%, propane -2.8%, Butane - 30-40% ; Iso Butane - 25%)

CHEMICAL CHANGE: (IRREVERSIBLE)

When a substance is heated the chemical composition can be changed and chemical properties would be varied. Those type of reactions are called chemical change.

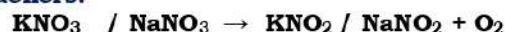
Eg:

Cryogenic Engines:

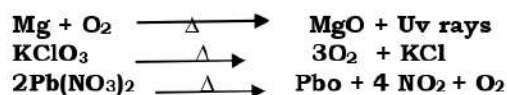


The major Exhaust from Cryogenic Engines is **Water vapor**.

(2) Crackers:



China crackers contain KClO_3 , NaNO_3 these are unstable. So easily produce oxygen and it has been banned in India since 1922.



The first man used ore - Malachite

The first man used metal – copper

(4) Formation of Rust:



Importance of $\text{Fe}_2\text{O}_3 \cdot \text{XH}_2\text{O}$

- | | |
|-----------------------------------|---|
| (a) Chemical formula of Rust | (e) The chemical name of rust is Ferric oxide |
| (b) It is a ferro magnet | (f) The responsible chemical for grey colour of cement |
| (c) it is also called as Hematite | (g) The responsible chemical for Red colour of soils |
| (d) The main ore of iron | (h) Fe_2O_3 is used to coat the tape of tape recorder |

CONTROLLING OF RUSTING (OR) CORROSION:

- (1) Galvanization
- (2) Electroplating method
- (3) Formation of Alloys
- (4) Coated with Graphene

GALVANIZATION: (FE WITH ZINC)

The process of spreading a layer of zinc on the Metal surface.

Spreading tin layer on iron is known as tinning.

Electroplating method:

In this mode, Covering cheap metal or material with a layer of an expensive metal.

In this method, cheap metal is made into cathode and a pure metal rod will be anode.

A solution of pure metal is used as electrolyte.

Eq: (a) Olympic God medal (92.5% Ag + 7.5% Au (or) 6 gm Gold) - Solid Gold 60 mm diameter, 3mm thick

(b) CD (compact disc) / DVD (Digital versatile disc) are coated with ferric oxide.

(c) Gramophone records etc.



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store

**FORMATION OF ALLOYS:**

An alloy is a combination of metals or metals combined with one or more other elements.

E.g.

1.	Steel	Fe + C + Mn	Utensils, machine parts
2.	Chromium Steel	Fe + C + Cr	Heavy machines
3.	Invar Steel	Fe + C + Ni	Pendulums, Tuning rods
4.	Stainless Steel	Fe + Cr + Ni	Surgical tools, Rupee coins
5.	Nichrome	Fe + Ni + Cr + Mn	Heater filaments
6.	Alnico	Fe + Al + Ni + Co	Permanent magnets
7.	Brass	Cu (70%) + Zn (30%)	Machine parts
8.	Bronze	Cu (85%) + Sn (15%)	First Man made Alloy
9.	Gun Metal	Cu + Zn + Sn	Weapons
10.	Aluminium Bronze (Roald Gold)	Cu (90%) + Al (10%)	Jewelry and photo frames
11.	Magnalium (Electron)	Mg + Al	Air Craft spare parts
12.	Solder metal	Sn + Pb	Fuse wire
13.	Duralumin	Al + Cu + Mn + Mg	Spare parts of Air craft.
14.	German silver	Ni+Cu+Zn	Spoons & Utensils
15.	Coinage mixture	Ag + Ni	To make silver coins
16.	PanchaLoha	Cu+Ag+Au+Fe+Zn/pb	
17.	Fire Alarms (Wood Metal)	Bi+Sn+ Pb + Cd	

Note: The percentage of Ag in German silver = Zero

The percentage of Lead in Lead Pencils = Zero

The percentage of Silver in Silver paints = Zero (Linseed oil + Aluminium powder)

The percentage of Silver in Quick Silver = Zero (Mercury)

Physical change	Chemical changes
1. Absorption (Attraction though out the Body)	1. Burning of fuels $C_{10}H_{20} \rightarrow CO + CO_2 + H_2O + NO$
2. Adsorption (only on surface)	2. Digestion of food Carbohydrate Gastec Juice Glycogen $\longrightarrow$
3. Crystallization (To attain Pure Compound)	3. Fermentation of Milk (or) curdling of milk Milk $\xrightarrow{\text{Yeast}}$ Curd (Lactose) (Lactic acid)
4. Sedimentation	4. Depletion of Ozone layer $O_3 \xrightarrow{CFC} O_2 + O$
5. Dissolving of glucose in water	5. Photo synthesis $n(CO_2) + n(H_2O) \longrightarrow C_6H_{12}O_6 + O_2$
6. Formation of Allotropes	6. Dissolving of sucrose in water

Why plants emit green colour?

Plants cells contain chloroplasts which have the pigment

Chlorophyll which absorbs Deep - Blue & Red Light

The rest of the sunlight spectrum is being reflected.

Colours impact on plants

Follow Chandan Logics on



Download Chandan Logics APP

Contact: 96 76 57 8793 / 94 94 55 8793

One STOP For ALL Competitive EXAMS





Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

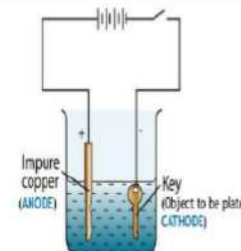
Contact: 96 76 57 8793

94 94 55 8793

Get it on Google Play

Download on the App Store

- Violet** - Enhances the colour, taste and aroma of plants
Blue - Growth rates of plants, start seedling
Green - Enhances chlorophyll production & used as a pigment for proper plant viewing
Red - Encourage stem growth, flowering, fruit production and chlorophyll production.
 12-14 hours per day needed.



WATER & ITS CONSTITUENTS

The Chemical name of water - Hydrogen Monoxide
 About 75% of the earth's surface consists of water.



Importance of water

- 1) Two third of the human body is water
- 2) Water Freezing point = 0°C
- 3) Water Boiling point = 100°C
- 4) It is also called as Universal solvent [Solution = Solute + Solvent]
- 4) Maximum density is $1.00\text{gm}/\text{cm}^3$ at 4°C
- 5) It has high dielectric constant.
- 6) The existence of Hydrogen Bonding is responsible for High values of specific heat, the latent heat of fusion and latent heat of Vaporization.

Why density of water is greater than density of Ice?

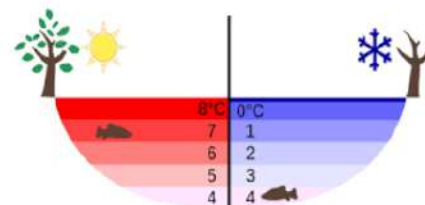
No. of Hydrogen Bonds in Ice = 4 (Hexagonal structure like Honey comb)

No. of Hydrogen Bonds in water = 4 (Tetrahedral)

Maximum number of cavities or gaps are existed in ice. so its density can be increased.

ANOMALOUS EXPANSION OF WATER

The anomalous expansion of water is an abnormal property of water whereby it expands instead of contracting when the temperature goes from 4°C to 0°C , and it becomes less dense. The density is maximum at 4 degree centigrade and decreases below that temperature.



Hybridization - sp^3

Shape - Angular (or) V-Shape

Bond angle - 104.5°

Ice melts slowly at High Altitudes explain why?

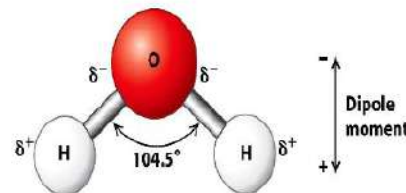
ANSWER: At High altitudes Atmospheric Pressures low. Hence it melts slowly.

Types of water:

- 1) Rain water
- 2) Acid Rain
- 3) Drinking Water (Mineral water, Ground water)
- 4) Sea water (Salt water)

RAIN WATER: $\text{pH} = 7$ (Neutral in nature)

- The purest form of water
- It acts as an Electrical insulator



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

Get it on Google Play

Download on the App Store

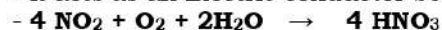


- It is Odorless, Colourless & Tasteless
- It is the source of Distilled water, Sterilized water.

ACID RAIN: $\text{pH} = 5.5$

- Acid rain is caused by a chemical reaction that begins when compounds like **sulfur dioxide** and **nitrogen oxides** are released into the air.

- it acts as an Electric conductor because of free ions

**ADVERSE EFFECT: (UNFAVORABLE CONDITIONS)**

- (1) It washes away nutrients needed for their growth of plants
- (2) It causes respiratory problems in Human beings and animals because of O_2 used by NO_2/SO_2 to form $\text{HNO}_3/\text{H}_2\text{SO}_4$ respectively.
- (3) It corrodes water pipes resulting in leaching of heavy metals like iron.
- (4) It damages the life of buildings.

DRINKING WATER: (PORTABLE WATER)

- 1) It is also called portable water
- 2) These are classified into two types
 - a) Mineral water
 - b) Ground water

Mineral water

- $\text{pH} > 7$
- It acts as an Electrical conductor (E.C) because of free ions
- Zn, Cu, Na, Mg etc. ions are added

Ground water

- $\text{pH} > 7$
- It acts as an Electrical conductor (E.C) because of free ions
- Calcium, magnesium Carbonates and Bicarbonates are present in it.

What is the best pH water to drink?

Ans: The U.S. Environmental Protection Agency recommends that the pH level of water sources should be at a pH measurement level between 6.5 to 8.5 on a scale that ranges from 0 to 14.

BRINE WATER / SEA WATER / SALT WATER

$\text{pH} = 7.9$ (Basic in nature)

- It acts as an Electric conductor due to presence of free ions
- The Most impure form of water is sea water
- The major impurity of sea water is NaCl.
- The average salinity of sea water is about 3.5% of the weight of sea water
- The Conc. of salt in sea water (Salinity) is about 3.5 parts per thousand comes from the dissolved salts in a cubic mile of sea water, the weight of the salt would be about 120 million tons,

→ The major abundant ion from sea water is Cl^- followed by Na^+

→ The metal extracted directly from sea water - Mg, Na

→ The nonmetal extracted directly from sea water - I₂

→ The main Source of Iodine is sea water

→ The best sink of atmospheric CO_2 is sea water

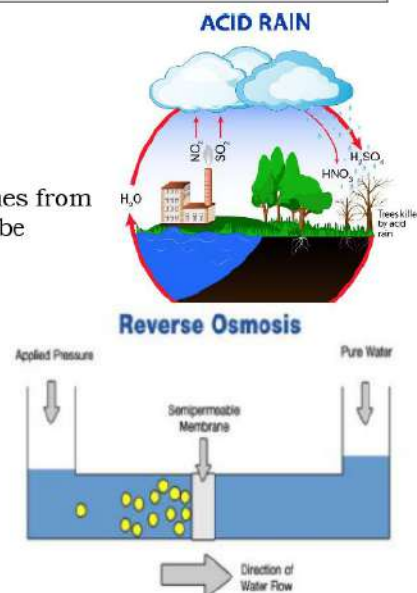
→ The important sinks for CO_2 are oceans, green plants.

METHODS OF CONVERSION OF SEA WATER INTO DRINKING WATER:

- (a) Distillation
- (b) Desalination
- (c) Reverse osmosis (Ro method)
- (d) Filtered with Graphene sheets

Reverse osmosis (RO):

It is a water purification process that uses a partially permeable membrane to separate ions, unwanted molecules and larger particles from drinking water under higher than osmotic pressure.





THE PURITY OF WATER DEPENDS ON FOLLOWING TERMS:

- (1) BOD (Biochemical Oxygen Demand)
- (2) COD (Chemical Oxygen Demand)
- (3) DO (Dissolved Oxygen)

1) BOD (BIOCHEMICAL OXYGEN DEMAND)

Biochemical oxygen demand (BOD) represents the amount of oxygen consumed by bacteria and other microorganisms while they decompose organic matter under aerobic (oxygen is present) conditions at a specified temperature (20°C).

Eg:	Clean water	-	1 PPM (BOD value)
	Normal water	-	5 PPM
	Water pollution	-	17 PPM
	Municipal sewage	-	100- 4000 PPM

Note: BOD increases – amount of oxygen decreases - Purity of water decreases

2) COD (CHEMICAL OXYGEN DEMAND)

The amount of oxygen required to oxidize organic substances present in polluted water is called COD.

Note: COD increases – amount of oxygen decreases - Purity of water decreases

3) DO (DISSOLVED OXYGEN)

The amount of oxygen present in water is called dissolved oxygen.

The Safest level Do in water is 15 PPM-9 PPM (PPM = parts per million)

E.g. Cold water - 10 PPM

Where as in air is about 2,00,000 PPM

Note:

- When dissolved oxygen becomes too low (less than 6PPM), fish and other aquatic organisms cannot survive.
- A moderate organic compound when decomposes in water can deplete the dissolved oxygen.
- The contamination of river water can be measured by knowing the quality of oxygen in the form of BOD, COD & DO values.

HARDNESS OF WATER

Based on hardness, Water can be classified into two types

- (1) **Soft water**
- (2) **Hard water**

(1) **Soft water:** It forms/gives lather readily with soap.
E.g. Ground water, river water

(2) **Hard water:** It cannot form lather easily with soap.
E.g. Sea water, salt water

**** The Hardness of water can be estimated with soap solution test.**

**** Consumption of Hard water causes - Kidney stones.**

**** The chemical name of kidney stones - Calcium oxalate calcium oxalate ($\text{CaC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$)**

The Hardness of Hard water can be classified into 2 types

- (1) Temporary Hardness
- (2) Permanent Hardness

TEMPORARY HARDNESS:

It is caused by calcium, magnesium Bicarbonates/Carbonates.

{ $\text{Ca}(\text{HCO}_3)_2$ = Calcium Bicarbonate; $\text{Mg}(\text{HCO}_3)_2$ - Magnesium Bicarbonate}

Biochemical oxygen demand





Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

Download on the

App Store

CaCO₃ - Calcium Carbonate; Mg CO₃ - Magnesium Carbonate}**Methods of Removal of Temporary Hardness:**

- (a) Boiling method (b) Clark's Method

(a) Boiling method:

During Boiling the soluble Mg (HCO₃)₂ & Ca(HCO₃)₂ can be changed to insoluble Magnesium Hydroxide and calcium Carbonate (CaCO₃).

**(b) Clark's method:**

Calculated amount of slaked lime is added to Hard water. It precipitates out calcium Carbonate and Magnesium Hydroxide which can be filtered off.

**PERMANENT HARDNESS:**

It causes due to calcium/magnesium sulphates (or) chlorides.

{Calcium Sulphate = CaSO₄ ; Magnesium Sulphates = MgSO₄

Calcium Chloride = CaCl₂ ; Magnesium Chloride = MgCl₂}

Methods of Removal of permanent Hardness:

- Washing Soda Method
- Calgon Method
- Ion Exchange Method (or) Permutt Method
- Synthetic Resin Method (Cation exchange resins; Anion exchange resins)
- Graphene

(a) **Washing Soda Method:** (Na₂CO₃·10H₂O - Sodium carbonate)

In this method soluble sulphates (or) chlorides can be converted into insoluble carbonates.



(b) **Calgon Method :** (Calgon - Sodium Hexa Meta Phosphate - Na₆ P₆ O₁₈)

When Hard water is passed over Calgon Ca⁺² & Mg⁺² is removed as follows. So Hard water is changed to soft water.



(3) **Ion Exchange (or) Permutt Method:** (Hydrated Sodium aluminium Silicates - Zeolite)

When Hard water is passed into zeolite Ca⁺² & Mg⁺² ions exchange Na⁺ ions.



{M = Ca⁺² / Mg⁺² ; Z = Aluminium silicates }

Exhausted permutate is regenerated by treating with Aqueous Sodium Chloride Solution.

(4) Synthetic Resin Method:

Resins are large organic molecules containing - Hydroxide (OH) / Carboxylic (COOH) Sulphate groups (SO₃H)

These are two types (1) Cation Exchange Resin

(2) Anion Exchange Resin

Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON

Google Play

Download on the

App Store

**Cation Exchange Resin:**

Cation of Resin can be exchanged by Ca^{+2} & Mg^{+2} ions.



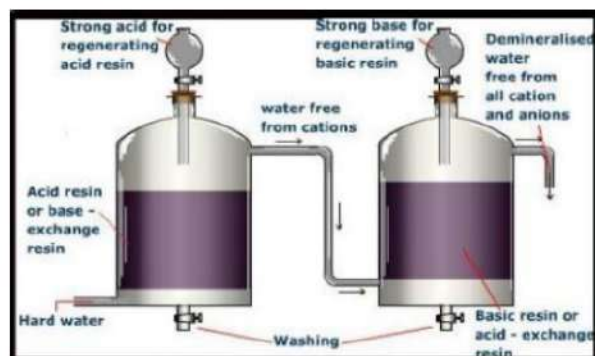
Resin can be regenerated by adding aqueous sulphuric acid.

Anion Exchange Resin:

NH_2 (or) OH^- (Anions)

Chlorides (or) Sulphates can be exchanged by Anion resin

$(\text{RNH}_2 + \text{H}_2\text{O})$

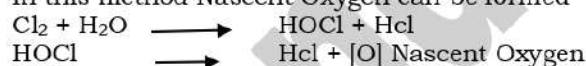
**Purification methods of drinking water:**

- (1) Chlorination
- (2) Bleaching powder
- (3) Ozonation
- (4) Ultra sounds
- (5) U.V-rays
- (6) Laser
- (7) Aeration
- (8) Activated charcoal
- (9) Graphene
- (10) R.O. method

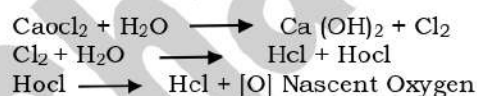
(1) Chlorination:

It means addition of chlorine gas (or) Bleaching powder.

In this method Nascent Oxygen can be formed

**(2) Bleaching Powder:** (CaOCl_2 - Calcium Hypo chlorite)

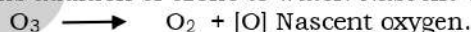
In this method Nascent Oxygen can be formed. It is used in drinking water as a Disinfectant.



Note: Chlorine exhibits two oxidation state in Bleaching Powder i.e. +1, -1

(3) Ozonation:

It means addition of ozone to water. Nascent Oxygen can be formed.



Nascent oxygen can be formed by using above three methods. It is responsible for killing of Germs in drinking water.

(4) Ultra Sounds:

Its Range is greater than 20000 Hertz (wavelength of 1.9cm or less)



**(5) UV - rays (Ultra violet rays)**

- Its Range - 10 nm to 400 nm (240 - 280 nm rays can be used to Dis infection, decontamination of surface and water)
- It is environmentally friendly method established more than 100 years ago and destroy microorganisms including bacteria, viruses effective against chlorine - resistant pathogens no detrimental taste or smell.

(6) Laser: Light implication by stimulated emission of Radiation.

(7) Aeration:

- The 1st major process at the water treatment plantation in municipalities .
- The process of increasing or maintaining the oxygen saturation of water in both natural and artificial environments.
- Aeration is the process of bringing water and air into close contact in order to remove dissolved gases, such as carbon dioxide and to oxidize dissolved material such as Iron. It can also be used to remove volatile organic chemicals (voc) in the water.

(8) Activated Charcoal:

it adsorbs colored particles easily

(9) Graphene**(10) R.O. (Reverse Osmosis Method)****Note:**

- The Drinking of Nitrate (NO_3^-) contaminated water causes - Blue baby syndrome (or) methemoglobinemia & it can be turned to stomach cancer.
- The safest level of Nitrate in drinking water = 45 mg/Lt.
- The percentage of Fluorine in drinking water is less than 1.5 PPM causes - Tooth decay (or) Dental cavities.
- The percentage of Fluorine is Greater than 4 PPM (or) 4 mg/Lt causes - Fluorosis.

The chemical name of soap is

S	-	Stearic acid
O	-	Oleic acid
A	-	And
P	-	Palmitic acids of sodium, potassium salts

Or Fatty acids of sodium potassium salts.

Saponification: (industrial Preparation Method of soap/detergent)

Alkaline hydrolysis of the fatty acid ester.

Oil/fat + KOH (Potassium Hydroxide) $\longrightarrow$ Toiler soap + Glycerol (Byproduct)

- The Reactivity of toilet soap can be measured with the percentage of TFM value.
- TFM (Total Fatty Matter) increases then Reactivity of soap increases
- The chemical name of Fat - Glycerol, Trimester of fatty Acids.
- The Quality of Detergent based on Height of the lather with Hard water.
- Height of the Lather increases then Reactivity of detergent also increases
- The chemical name of Detergent is sodium salts of benzoic sulphates.

Note: some important oil or fats:

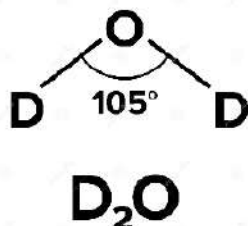
Oleic acid - $\text{C}_{17}\text{H}_{33}\text{COOH}$ - Unsaturated
 Linolic acid - $\text{C}_{17}\text{H}_{29}\text{COOH}$ - Unsaturated - Lin seed oil
 Stearic acid - $\text{C}_{17}\text{H}_{35}\text{COOH}$ - Unsaturated - Fat/ Butter
 Lauric acid - $\text{C}_{11}\text{H}_{23}\text{COOH}$ - Saturated - Coconut oil Butter.



**DIFFERENT TYPES OF FATTY ACID SALTS:**

Fatty Acids	Uses
K ⁺ Salts	Toilet soap
Na ⁺ Salts	Washing soap
Mg ⁺²	Talcum powder
Ca ⁺² , Al ⁺³	Water proof cloth / paper
Li ⁺	Grease
Triethanol Ammonium	Dry cleaning

DRY CLEANING: The Chemicals used is Kerosene, Benzene, Alcohol, Carbon tetrachloride etc.

**HEAVY WATER** D₂O (D = ²H)

- **D₂O** - Chemical name = Deuterium oxide
- Deuterium is a Non-radioactive Isotope of Hydrogen
- The percentage of D₂O in ordinary water 1/6000
- It was discovered by Dr. Urey
- 1ml of D₂O produced from 30 Lt of water by Exhaustive Electrolysis (E.F)

Applications:

- (1) It is used as a moderator in Nuclear Reactors
- (2) It is used as a coolant.

D₂O is manufactured by HWB (Heavy Water Board) in India.

HWB Plants location: 1) Nangal 2) Rourkela 3) Trombay 4) Nayeli

HYDROGEN PEROXIDE: (H₂O₂)**(1) Open book structure****(2) Dia magnetic in Nature**

3) Pure anhydrous H₂O₂ is a syrupy liquid

(4) It is colorless

(5) It is soluble in water, alcohol & ether forms a hydrate H₂O₂ · H₂O

(6) Bitter in taste

(7) Pure H₂O₂ - Weak Acidic

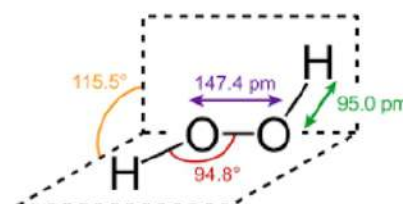
(8) Aqueous H₂O₂ - Neutral

(9) It is unstable & decomposes slowly on exposure to light as water and oxygen.



To prevent H₂O₂ decomposition following compounds can be used.

- (a) Area
- (b) Acetanilide
- (c) Pyrophosphate
- (d) Phosphoric acid

**STORAGE OF H₂O₂:**

(1) Wax coated Bottles (Rough surface may enhance the decomposition of H₂O₂)

(2) It is kept away from dust because dust can induce explosive decomposition of H₂O₂.

APPLICATIONS:

(1) It is used for restoring color of old oil lead painting



(2) An Antichlor (remove traces of chlorine)

(3) Rocket propellant (40% H₂O₂ + Hydrogen + Alcohol)

(4) As a preservative for milk & wine

(5) In daily life as a hair bleach

(H₂O₂ + NH₃ - Auricome) as a mild disinfectant

(6) An Antiseptic agent and sold in the market as perhydrol (30% H₂O₂)





Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**Contact: **96 76 57 8793****94 94 55 8793**GET IT ON
Google PlayDownload on the
App Store

Elements and its applications

ELEMENT: A basic form of matter that cannot be broken down into simpler substances by Chemical properties.

E.g. P₄, S₈, C₆₀, Diamond, Graphite

** The term "element" was first used by "Robert Boyle"

** The Definition of element was proposed by "Lavoisier"

** Total number of elements in modern periodic table = 118

** The total number of discovered elements in modern periodic table = 118

Based on physical state Elements can be classified into three types

- 1) Solids
- 2) Liquids
- 3) gases

*** Mercury is the only metal that is Liquid at room temperature

*** Bromine is the only nonmetal that is liquid at room temperature.

Based on Nature Elements can be classified into three types

- 1) Metals (95)
- 2) Nonmetals (17)
- 3) Semi Metals (6)

MOST ABUNDANCE ELEMENTS

	Universe	Atmosphere	Human Body	Earth Crust
1 st	Hydrogen (75%)	Nitrogen (78%)	Oxygen (65%)	Oxygen (46.1%)
2 nd	Helium (21-23%)	Oxygen (21%)	Carbon (18.5%)	Silicon (28.2%)
		Argon = 0.93%	Hydrogen (9.5%) Nitrogen (3.2%) Calcium (1.5%)	Metals Aluminium (8.2%) Fe (5.6%)

BITS:

- (1) The permanent gases of the universe - Hydrogen, Helium
- (2) The most abundance noble gas in the atmosphere - Argon
- (3) The percentage of Nitrogen & Oxygen in the atmosphere = 4:1 (Volume)
- (4) Primary elements of human body Oxygen, Carbon, Hydrogen
- (5) The Most abundance metal in human body - Calcium
- (6) The Most abundance Metal in the earth crust - "Al" (Aluminium)

HYDROGEN (ATOMIC NUMBER - 1 ; MASS 1 AMU)

- (1) Hydro means "water"; gen means - produce / forming. So, hydrogen means water producing element
- (2) Discovered by Henry Cavendish (1766)
- (3) Most abundance element in the universe/stars/sun
- (4) The lightest gaseous element.
- (5) The gas with Highest rate of diffusion
- (6) The velocity of sound maximum in H₂ gas (1270 m/sec)
- (7) The most common element in all acids

E.g.	Hydrochloric acid	-	Hcl
	Nitric acid	-	HNO₃
	Sulphuric acid	-	H₂ SO₄
	Formic acid	-	HCOOH

Follow **Chandan Logics** onDownload **Chandan Logics APP**Contact: **96 76 57 8793 / 94 94 55 8793**One **STOP** For **ALL Competitive EXAMS**GET IT ON
Google PlayDownload on the
App Store



- (8) It has one electron, one proton and no neutron in the podium isotope.
- (9) It has highest calorific value in the trade name of cryogenic fuel
 e.g. Liquid $H_2 = 141970 \text{ KJ/Kg}$
 Petrol = 48000 KJ/Kg
 Diesel = 44800 KJ/Kg
 Kerosene = 46200 KJ/Kg
 Natural gas = 55000 KJ/Kg
- (10) The gas burns with Blue colour flame and makes a pop sound.
- (11) Oxy-Hydrogen blow torches used for cutting and welding process (2400°C)
- (12) $\text{Metal} + \text{Acid} \longrightarrow H_2 \text{ gas}$
 E.g. Li/Cs/Rb/K/Ba/Sr/Ca/Na/Mg/Al/Mn/Cr/Cd/Co (metal)
- (13) $\text{Metal} + \text{Acid} \longrightarrow H_2 \text{ gas cannot be evolved}$
 e.g. Ag/Cu/Pt/Hg/Au/ CO^{+3} (Metals)
- (14) $\text{Metal} + \text{Base} \longrightarrow H_2 \text{ gas}$
 e.g. Na/Mg + NH_3
 Zn/Al + NaOH
- (15) $\text{Metal} + \text{Cold } H_2O \longrightarrow H_2 + \text{Metal oxides}$
 Eg: Na/Ca/Ba (Metals)
- (16) $\text{Metal} + H_2O \text{ (Hot)} \longrightarrow H_2 + \text{Metal oxides}$
 Eg: Zn/Al/Mg
- (17) $\text{Metal} + \text{steam} \longrightarrow H_2 + \text{Metal oxides (Lane's process)}$
 Eg: Fe/Mn/Co/Ni/Cr/Sn
- (18) $\text{Nascent Hydrogen [H]} + \text{Colour substance} \longrightarrow \text{colourless}$
- (19) **Hydronification of oils (Discovered by Berzelius)**
 Vegetable oil $[C=C] + H_2 \text{ gas} \xrightarrow{Ni (473k)}$ Dalda / Vanaspati $[C-C]$
 Unsaturated Hydro Carbons Saturated Hydro Carbons
- (20) **Artificial petrol (Berzelius)**
 Oil + $H_2 \text{ gas}$ + Charcoal powder $\xrightarrow{Ni/Pt 450^\circ\text{C}}$ Liq fuel + Liq gas
- (21) **Haber's process:**
 An Industrial process for the preparation of ammonia
 $3H_2 + N_2 \xrightarrow[Mo]{Fe} 2NH_3$
 Catalyst - Fe ; Premotor- Mo (Molybdenum)



Ortho & Para Hydrogen

A Molecule of Hydrogen contains two atoms. Depending upon the direction of the spin of the Nuclei, the Hydrogen is of two types.

- (1) Para Hydrogen - Spins of both nuclei are in the same direction.



Chandan Logics

Chandan Logics

Aspire to Inspire

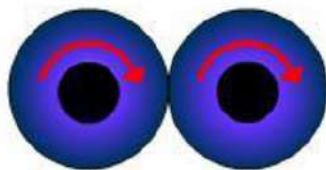
Download **Chandan Logics APP**

Contact: 96 76 57 8793

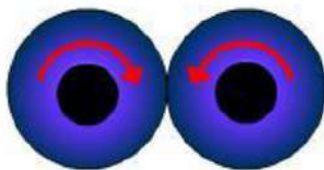
94 94 55 8793

GET IT ON Google Play

Download on the App Store



Ortho hydrogen

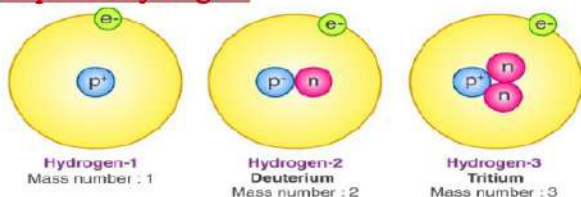


para hydrogen

(2) **Ortho Hydrogen** - spins of both Nuclei are in the opposite direction.

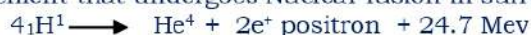
- Para Hydrogen is more stable at zero kelvin
- The ratio of ortho: Para hydrogen at room temp. (or) very high temperature = 3:1
- Ortho: Para Hydrogen (at Liquid air temperature) 1:1

Isotopes of Hydrogen:



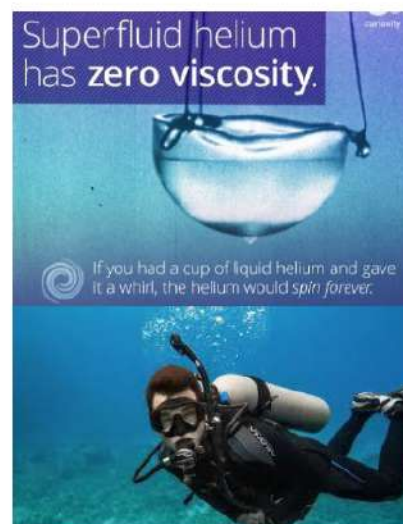
The lightest Radioactive element is Tritium

The element that undergoes Nuclear fusion in sun's chromosphere/ stars / Hydrogen Bomb



Helium (He):

- In Latin Helios means sun
- It can be stored in Brown colour cylinders
- The Lightest inert gas
- A Mono atomic gas
- It was discovered by French Astronomer (1868) "Pierre Janssen" during the solar eclipse at Guntur.
- Specific Heat ratio $c_p/c_v = 1.66$
- It is used in filling of balloons for meteorological observations.
- Liquid Helium is used to provide low temperature for the storage of super conducting materials.
- It has least electro negativity.
- The element with lowest boiling point = -268°C (or) 4kelvin
- The Permanent gases of this universe - Hydrogen, Helium
- Most of the Helium in the universe can be generated during the time of big bang collision.
- The element with Highest Ionization Potential in the nature.
- Liquid Helium is regarded as most volatile of all liquids
- "He" is the only noble gas without 'P'- orbital electrons.
- α - particle (Alpha) Possessed Helium nucleus (${}_2\text{He}^4$)
- The chemical composition in Scuba tank (Longer limit) is Nitrox - Enriched air [22-40% O_2 + 60% N_2]
- Patients with difficult berating (Asthma) - 70% He + 30% O_2
- Trimix - (N_2 + O_2 + He) - Depth limit increases (or) dive time increases.
- Ease of liquification of noble gases increases from Helium to xenon due to increase of Vander waal's forces. $\text{He} < \text{Ne} < \text{Ar} < \text{Kr} < \text{Xe}$
- If Helium cooled to 2.2k at 1 atmospheric pressure it changes to liquid known as He-II (very low viscosity). It flows upwards instead of flowing downwards.



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON Google Play

Download on the App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

- At 4.2k Helium becomes liquid and is called Helium-I on further cooling to 2.2k He (II) is formed because of its low viscosity.

Lithium: (Li)

- In Greek Letho's Means - Stone
- The Lightest Metal
- The Metal with Highest Hydrated energy
- It can be used to make rechargeable batteries like Mobile, Computers, Laptops etc.,
- Hydrogen fuel can be stored in the form of Lithium Hydrate
- Most Hygroscopic material - Lithium chloride
- Lithium derivatives can be used in the preparation of Grease.
- Lithium - Aluminium Alloy can be used to make Aircraft /Bicycle Frames.
- Dow Metal = Mg + Al - It is used to make spare parts of motor vehicle.

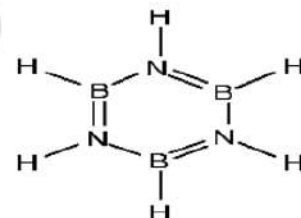


Beryllium (Be)

- Beryllium oxide (BeO) can be used as a moderator in Nuclear reactors.
- Beryl is the main ore of Beryllium
- Beryl $\text{Be}_3 \text{Al}_2 (\text{SiO}_3)_6$.

Boron: (B)

- ➔ Isotopes of Boron ${}^5\text{B}^{10}$ (19%); ${}^5\text{B}^{11}$ (81%)
- ➔ Borax powder / Tincal / Kernite - $\text{Na}_2 \text{B}_4 \text{O}_7 \cdot 10\text{H}_2\text{O}$
- ➔ The number hydrated molecules in Borax - 10
- ➔ Boric acid - H_3BO_3 - Mono Basic compound
- ➔ Boric acid can be used as Antiseptic agent.
- ➔ High purity 'Boron' can be used as a semiconductor in the place of 'Ge' and 'Silicon'
- ➔ ${}^5\text{B}^{10}$ Isotope, Boron carbide and cadmium are used in the nuclear Industry as protective shields and control rods ${}^5\text{B}^{10} + {}^1_0\text{n}^1 \rightarrow {}^5\text{B}^{11}$
- ➔ ${}^5\text{B}^{10}$ Isotope has High ability to absorb neutrons.
- ➔ 'B' is essential micro element for the Healthy growth of plants.
- ➔ Boron fibers are used in making Bullet proof vest and Light composite material for Air craft.
- ➔ White graphite --- Boron Nitride $(\text{BN})_x$
- ➔ Inorganic Benzene - Borazole $(\text{B}_3 \text{N}_3 \text{H}_6)$

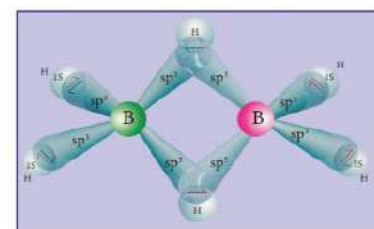


Diborane -- $\text{B}_2 \text{H}_6$

It has 3 center - 2 electrons system

It is also called Banana Bond / H-bridge bond / Tau bond

- ➔ The match stick burns completely till the end because the stick is dipped in Borax solution.



Borax Brad Test (Blister test)

- ➔ It was introduced by Berzelius

The identification test of Basic radicals in quantitative analysis.

Boric Anhydride + Metal Oxide → Metal Meta borate



Cobalt meta borate	-	Blue bed
Copper meta borate	-	Green
Iron Meta borate	-	Yellowish brown
Chromium Meta borate	-	Green

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793



Manganese meta borate	-	Violet
Nickel meta borate	-	Violet

Carbon (C) :

- The king of elements
- Carbon can form more than 50 lakhs of compounds whereas all others produce only one lakh compounds.
- The element with Highest catenation property.
- The element with Highest atomicity C_{60} , C_{70} , C_{80} .
- The element that exists in both Hard and Soft forms is Carbon (Diamond; Graphite)
- The compound used for Lubrication in eyes is cellulose acetate.
- The Chemical name of Artificial silk - Rayon.
- The Hardest element
- The most abundance carbon compound in nature is cellulose ($C_6H_{10}O_5$)



(a) Water gas/syngas/blue gas - $CO + H_2$ - Industrial Fuel

(b) Producer gas - CO (32%) + N_2 (64%) + O_2 (3%) - Industrial Fuel (Steel & glass Industry)

(c) Semi Producer gas - CO (30%) + N_2 (50%) + H_2 (10%) + CH_4 (10%) - Fuel in steel industry

(d) Coal gas - CH_4 (35%) + CO (10%) + H_2 (50%) - Domestic & Industrial fuel in Metal Industry.

Bhopal Disaster:

- 1984 - Dec - 3rd
- Responsible chemical: CH_3NCO - MIC gas - Methyl Isocyanate
- Name of the industry: UCIL - Union Carbide India Limited
- Type of chemical : Pesticide plant in Bhopal (Madhya Pradesh)



Carbon dating Method:

- This method was invented by Prof. Libby
- It depends on $^{14}C / ^{12}C$ ratio
- ^{14}C is a radioactive element (Half life period - 5770 years)
- This method is used to find out the age of Ancient organism, Biological spices.
- It is valid up to the age of an Antique is below 30,000 years.



Nitrogen:

- It can be stored in Black Cylinders
- Discovered by Rutherford
- Boiling point = $-196^\circ C$
- Animal sperm can be preserved in Liquid Nitrogen
- In Electric bulbs, chips packets and pickle jars, for getting inert atmosphere the gas filled is N_2 .
- The chemical which is used in High temperature thermometers - N_2 gas
- The gas Liberated due to decomposition of dead animals - N_2 gas
- The essential elements for the growth of plants is Nitrogen, Phosphorus, Potassium (N,P,K)
- The major pollutants from Supersonic Jet planes - Nitric Oxide
- The frequent rains increase the content of Nitrogen in soil
- The first man made Bio organic compound - urea (or) Carbamide (NH_2CONH_2)
- The responsible chemical for pungent odor of Toilets is Ammonia (NH_3)



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**





Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON Google Play

Download on the App Store

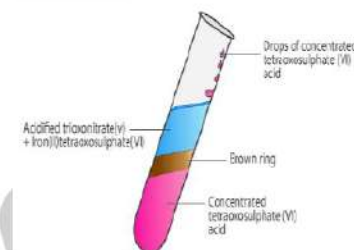
- Ammonia can be used as a refrigerating agent in Heavy Refrigerator
- The irritant red haze of the eyes in the traffic and congested places is due to Nitrogen dioxide.

Brown ring test:

- The familiar test to detect Nitrate (NO_3^-) ion.
- The compound formed is ferrous Nitrososulphate $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$ (or) $\text{FeSO}_4 \cdot \text{NO}$
- Iron oxidation state is +1 in Brown ring test.
- Excess amount of Nitrate (NO_3^-) ions in drinking water causes - Blue Baby syndrome (or) Methemoglobinemia
- Plants absorb Nitrogen in the form of Nitrates only.

Nitrogen fixation:

Transformation of atmospheric Nitrogen by bacteria (rhizobium) in the roots of plants (Leguminous) of useful Nitrogenous compounds (NO_3^-).



Atmospheric Nitrogen $\xrightarrow{\text{Rhizobium Bacteria}}$ Nitrates (NO_3^-)

N_2 $\xrightarrow{\text{Nitrogenase Enzyme}}$ Ammonia

Nitrogenous compounds $\xrightarrow{\text{Denitrifying Bacteria}}$ Nitrogen

- Nitrogen fixation (N_2 to NH_3 / NH_4^+ or NO_3^-)
- Nitrification (NH_3 to NO_3^-)
- Assimilation (Incorporation of NH_3 and NO_3^- into biological tissues)
- Ammonification (organic nitrogen compounds to NH_3)
- Denitrification (NO_3^- to N_2)

Oxygen:

- In Greek "oxy" means "acid"; "Gen" means - produce.
- It can be stored in white colour cylinders.
- Discovered by Scheele (1772) by heating Hgo = Mercuric oxide)
- Joseph Priestly was also prepared oxygen by the decomposition of KNO_3 and mercuric oxide (1774 - August)
- The name of this gas was proposed by Lavoisier
- The most abundance element in Human body - 65%
- The most abundance element in Earth crust - 46%
- The percentage of oxygen in Silicon - 56%
- The percentage of oxygen in water - 85%
- The gas needed for the combustion of fuel is O_2 .
- Oxy acetylene flame - 3200°C
- Oxy Hydrogen flame - 2400°C
- Oxygen can be prepared by the decomposition of potassium chlorate (KClO_3) in laboratories



Roasting: The process of strong heating of the ore in presence of air (or) oxygen. This process is suitable for sulphide ores.



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON Google Play

Download on the App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793



Calcination: The process of strong heating of the ore in absence of air (or) in limited supply of air but below its fusion temperature is called calcination.

Eg:



Combustion: It is a chemical process in which a substance reacts rapidly with oxygen and gives off Heat.

Allotropes of Oxygen

- 1) O_2 (Para Magnet)
- 2) O_3 (Dia magnet)

Ozone: (O_3)

- Atomicity - 3
- It has Both oxidizing and reducing nature
- Rotten fish smell
- Ozone layer is located in strato sphere - 23 km height of altitude from the sea level
- Major role of ozone layer: It filters UV rays only
- Thickness - 300 Dobson Units.

Depletion of Ozone Layer:

Caused by

- (a) Choro Fluor Carbons (EFC)
- (b) Carbon dioxide
- (c) Chlorine
- (d) Bromine
- (e) Methane
- (f) Nitric Oxide
- (g) PAN - Para Acetyl Nitrate
- (h) Photo chemical smog (Nitrogen Oxides)

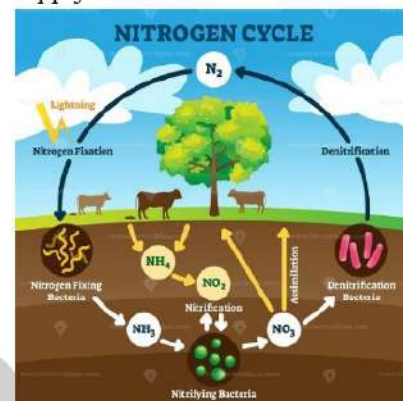
Montreal Protocol: The first international agreement to protect ozone layer from its depletion gases.

16th - September - 1987

Alternative for CFC:

- (a) HFC - Hydro Fluoro Carbons
- (b) Ammonia
- (c) Hydro Chloro Fluoro Carbons
- (d) Methyl / Cyclohexane / Butane & Propane

	Temperature	Types of gases present
Tropo sphere	15 ⁰ c to -56 ⁰ c	O_2 , N_2 , CO_2 , H_2O
Strato sphere	-56 ⁰ c to -2 ⁰ c	O_3 , N_2 , O_2
Meso sphere	-2 ⁰ c to -92 ⁰ c	O_2^+ , NO^+
Thermo sphere	-92 ⁰ c to 1200 ⁰ c	O_2^+ , O^+ , NO^+



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**





Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

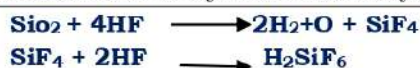
94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

Fluorine: (Yellow colour)

- Super halogen
- The most reactive non metal
- The element with highest electronegativity (4)
- Halogens ("Halo - sea; gen = produce)
Eg: F, Cl, Br, I
- It can be stored in Teflon coated cylinders (or) Gutta-percha (Trans form of 2 -Methyl 1,3 butadiene)
- It acts an oxidizing agent
- Glass dissolves in Hydrofluoric salicylic acid (H_2SiF_6)



- 'HF' is used for etching (or) making of glass
- Freon - Fluoro chloral carbon (Refrigerant - CCl_2F_2)
- Teflon - Polymeric tetra Fluoro ethylene
- Teflon can be used as an anti-corrosive plastic
- Fluorine can be used to separate U^{235} and U^{238} in the form of UF_6 gases based on their rates of diffusion

$$r_1/r_2 = \sqrt{d_2/d_1} = \sqrt{m_2/m_1} \quad \begin{aligned}r &= \text{rate of diffusion} \\ d &= \text{density of gases} \\ m &= \text{molar mass}\end{aligned}$$

- Sodium Fluoride (NaF) & cryolite Na_3AlF_6 can be used as insecticides.
- Fluorine can be used as rocket fuel



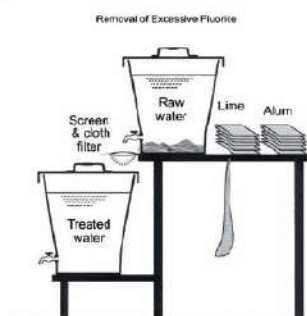
- SF_6 is used as faced insulator in high voltage electricity.
- The element needed for the growth of enamel layer is Fluorine.
- The chemical name of enamel layer is Calcium hydroxyl apatite Fluor apatite $\text{Ca}_3(\text{PO}_4)_2 \cdot \text{CaF}_2$
- The amount of fluorine in drinking water is below 1.5 PPM causes dental decay / tooth decay.
- To avoid dental decay, the doctors prescribed Fluoride tooth paste.
- The amount of fluorine is greater than 4 PPM or 4 Mg / Lt in drinking water causes fluorosis (Bones become weak and yellow plaque on teeth)



Nalgonda Fluorination Method:

- The cheapest method to remove excess of fluorine from soft water
- **Developed by NEERI - National Environmental Engineering Research Institute (Nagpur)**

Chemicals : Bleaching powder
Lime water
Alum



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

Neon: (Ne)

- Latin name – new gas
- It can be red colour signal light / red colour sign boards / red colour railway signal lights etc.

Sodium: (Na)

- Latin name - Natrium
 - Discovered by Humphrey Davy
 - The most abundant metal in sea water (30.6%)
 - It can be stored under kerosene
 - Liquid sodium has been used as a coolant for nuclear reactors
 - Sodium vapors are used in street lights and produce a brilliant yellow light.
 - When sodium reacts with water, Nascent Hydrogen can be formed.
- $$\text{Na} + \text{H}_2\text{O} \longrightarrow \text{NaOH} + [\text{H}]$$
- It can be used as a catalyst in the manufacture of Artificial rubber.
 - Soda Lime - $\text{NaOH} + \text{CaO}$
 - Washing soda - sodium carbonate $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
 - Baking soda / Eating soda - sodium Bicarbonate - NaHCO_3
 - Soda ash - Na_2SiO_3
 - The Metal which can extract from Borax.

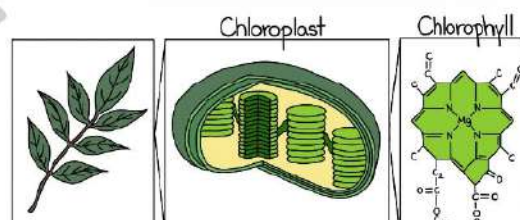


Sodium Vapour Lamp

Magnesium: (Mg)

- The second most abundant Metal in seal water
- The abundant metal in chlorophyll
- The chemical constituents of pearls - MgCO_3 CaCO_3
- Talc powder - MgSiO_3 - Magnesium silicates
- Purgative for children's - Mg(OH)_2 - Magnesium Hydroxide
- Purgative for Adults - $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ - Epsom salt
- The chemical ingredient of tooth paste - CaCO_3
- Ores of Mg: -

Magnesite	-	MgCO_3
Dolomite	-	$\text{MgCO}_3, \text{CaCO}_3$
Carnallite	-	$\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
Epsom salt	-	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$



Aluminum : (Al)

- The most abundant metal in earth crust (8.3%)
- The percentage of silver in silver paints is zero
- The chemical composition of silver paints Aluminum powder + Linseed oil.
- The catalyst used in Friedel craft's reaction is Aluminum chloride (AlCl_3)
- The main ingredient present in ornamental stones (ruby (or) sapphires) - Aluminum oxide
- The main ore of Aluminum - Bauxite
- The impurities present in Red Bauxite - Fe_2O_3
- The impurities present in White Bauxite - SiO_2 (Silicon)
- The self-productive metal is aluminum



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON Google Play

Download on the App Store

- Nitric acid (HNO_3) can be transported in an aluminum container because the reaction between Aluminum and Nitric acid forms a thin layer of oxide on the aluminum surface which makes it passive.

Ores of 'Al' :	(a)	Corundum	- Al_2O_3
	(b)	Dyspore	- $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$
	(c)	Bauxite	- $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
	(d)	Cryolite	- $\text{Na}_3\text{Al F}_6$



Silicon: (Si)

- The Best Semi Metal
- The second most Abundant element in the earth crust
- Silicon & Germanium, are used in transistors as Metalloids.
- The purest form of silica is quartz.
- Solids which conduct electricity at higher temperature but not at lower temperature are called - semiconductors.
- It can be used to make semi-conductors

Types of Semiconductor Devices



Semi-Conductors

P - type

Electrical carriers - Holes
Silicon + 3rd Group
Eg: Si/Ge + Gallium

n - type

Electrical carriers - Electrons
Silicon + 5th group
Eg: Si/Ge + Arsenic

Phosphorus: (P)

- Highly Reactive phosphorus allotrope
- It can be stored in water
- Rat poison (or) Rodenticide - Zinc phosphide (Zn_3P_2)
- When white phosphorus taken out into air burns to produce P_2O_5 (Phosphorus Penta oxide). It is called as "phosphorescence".
- Garlic odor - P_2O_5 , white 'P', Sulphur
- The chemical composition of "Holmes's signals" calcium phosphide + Calcium Carbide.

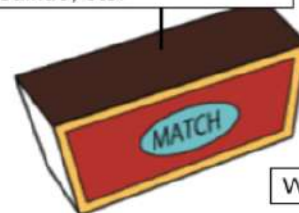
Red phosphorus:

- It can be used in Match Box Industry.

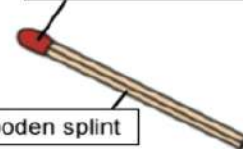


Side chemical:
red phosphorus, antimony
sulfide, etc.

Head chemical:
potassium
chlorate,
glass powder, etc.



Wooden splint



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON Google Play

Download on the App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

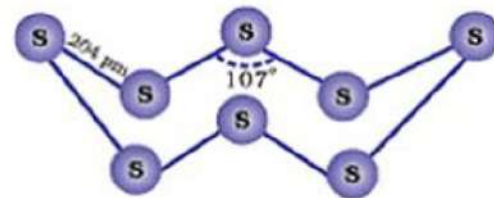
94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

Sulphur: (S)

- Molecular formula - S_8 ; Atomicity = 8
- Crown shape (or) Boat Shape Element
- It is also called Brim Stone (Burning stone)
- Rotten egg smell - H_2S (Hydrogen Sulphide)
- Acid rains mainly caused by Sulphur dioxide
- To identify the leakage of LPG cylinders the gas added is Methyl Mercaptan (CH_3SH)



Vulcanization of Rubber:

→ **Discovered by 'Charles Good year'**

- The process of heating Natural rubber with Sulphur at 373 - 415K to improve its properties is called Vulcanization.
- Allotropes of 'S' --- (a) Rhombic (α -Sulphur)
(b) Mono clinic (beta - Sulphur)
(c) Plastic (gamma - Sulphur)

Most stable one - rhombic Sulphur



The transition temperature of Sulphur is 95.6°C

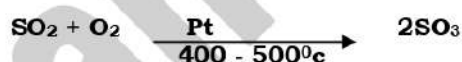
Rhombic Sulphur $\xrightleftharpoons{< 95.6^\circ\text{C}}$ **mono - Sulphur**

Preparation Methods of Sulphuric acid (H_2SO_4)

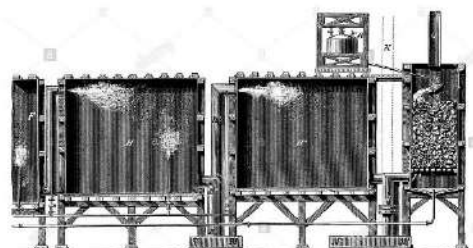
(1) **Contract process**



(2) **Lead chamber process**

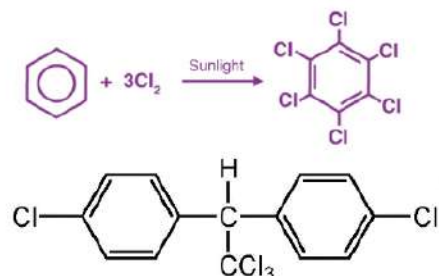


- Oleum - $H_2S_2O_7$ - fuming Sulphuric acid.



Chlorine (Cl)

- The element with highest electron affinity
- The most commonly used bleaching agent
- DDT - Dichloro diphenyl trichloro Ethane (Insecticide)
- BHC - Benzene hexa Chloride - Gammaxene (Lindane) (Insecticide)
- Mosquitoes Killer - Pyrethrin



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

Chemical weapons

- 1st world war - (a) Mustard gas
(b) Phosgene
- 2nd world war - (a) Lewisite
(b) Tcargas / Chloropicrin ($\text{Ccl}_3 \text{NO}_2$)

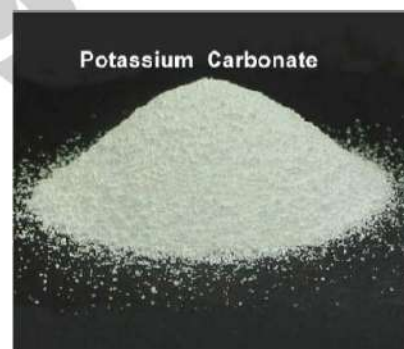
Argon (Ar) (Lazy gas)

- The cheapest Noble gas
- The most abundant Noble gas in the earth atmosphere
- The most widely used Noble gas
- White colour street lights - Mercuric vapors + Argon
- It can be used in Geiger counter tube to identify the radioactive rays like alpha, beta and gamma rays.



Potassium: (K)

- Latin Name - Kalium
- it can be stored under kerosene
- Pearl ash - potassium carbonate (K_2CO_3)
- Red medicine - potassium permanganate - antiseptic agent
- Liquid potassium can be used heat carrier in High temperature thermo meters.
- To enhance the hardness of glass the substance added is potassium chloride.
- **Ores**
 - **potassium chloride**
 - **potassium carbonate**
 - **potassium nitrate**



Calcium (Ca)

- The most abundant metal in Human body (1.5%)
- The element with least electron affinity.
- The chemical name of kidney stones - calcium oxalate
- It is a vital mineral to stabilize Blood pressure, build strong bones and teeth.
- Dehydrating agent - CaCl_2
- The massive amount of calcium present in the form of Human Body - Calcium phosphate.
- The chemical which is used to Artificial ripening of fruits - Acetylene (or) Calcium carbide.
- The chemical used for dispelling mosquitoes - Mono calcium Hydrogen Phosphate.
- The commercial name of Calcium Hydride - Hydrolith Osteoporosis / Hypocalcemia
- Lime mortar - Quicklime (CaO) + silica (SiO_2) + H_2O
- Cement mortar - Cement + Sand + Water
- Superphosphate of lime - $\text{Ca} [\text{H}_2 (\text{PO}_4)]_2 \cdot \text{H}_2\text{O} + 2\text{Ca} \cdot \text{SO}_4 \cdot 2\text{H}_2\text{O}$



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

- **Ores**
- Gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
 - Plaster of Paris = $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
 - Calcium carbonate CaCO_3 (Lime stone)

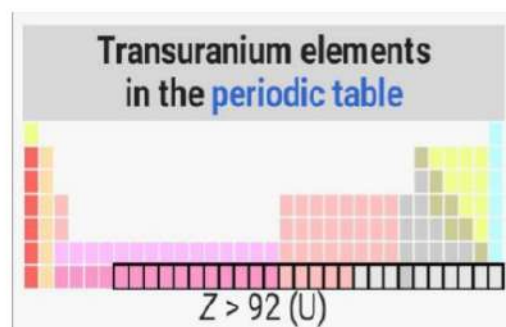


- 1) Plaster of Paris (POP) is used as bone cement
- 2) Chemical name of POP is hemi Hydrated calcium sulphate

Classification of Elements and Periodicity in Properties

- The total number of elements in periodic table - 118
- The total number of discovered elements so far - 118
- The four newly discovered elements -

Atomic number	proposed name	Symbol
113	Nihonium	Nh
115	Moscovium	Mc
117	Tennessine	Ts
118	Oganesson	Og



- Elements coming after 92 atomic number are known as "Trans uranic Elements" (Radioactive)

Dobereiner Law of Triads

- Elements were first classified by Dobereiner.
- "Triads"-- A certain group of elements with similar properties.
- The atomic weight of the Middle element was approximately the Arithmetic mean of the other two.
- $2^{\text{nd}} \text{ element weight} = \frac{(1^{\text{st}} + 3^{\text{rd}}) \text{ Weight}}{2}$

Triads	(a)	Li	Na	K	'Na' Mass = $\frac{7+39}{2}$
		7	23	39	
	(b)	Cl	Br	I	Mean = 81.25
		35.5	80	127	
	(c)	Ca	Sr	Ba	Mean = 88.5
		40	88	137	

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store

**Demerits**

→ Some triads possess nearly equal atomic masses

Eg: Fe, Co, Ni ; OS, Ir, Pt

DOBEREINER'S TRIADS

	ELEMENTS	SYMBOL	ATOMIC MASS
1	Lithium	Li	6.9
	Sodium	Na	23
	potassium	K	39
2	Calcium	Ca	40.1
	Strontium	Sr	87.6
	Barium	Ba	137.3
3	Chlorine	Cl	35.5
	Bromine	Br	79.9
	Iodine	I	126.9
4	Sulphur	S	32
	Selenium	Se	79
	Tellurium	Te	128

De-chaan courtois:

The known elements arranged in order of increasing atomic weight and made cylindrical table of elements.

Newland octaves:

The Law of octaves "If the known elements are arranged in the increasing order of their atomic weight, the 8th elements had properties similar to those of first element as the eight one of octaves.

Eg: (1) (2) (3) (4) (5) (6) (7)
 Li Be B C N O F
 (8) (9) (10) (11) (12) (13) (14)
 Na Mg Al Si P S Cl
 (15) (16)
 K Ca

Demerit:

This law is true only for the elements up to calcium.

Lothar meyer:

Lothar meyer plotted the '**Physical Properties**' such as atomic volume, melting point & Boiling point against '**atomic weight**' and obtained a periodically repeated pattern.

$$\text{Atomic weight} = \frac{\text{Molecular weight}}{\text{density}}$$

MENDELEEV'S CLASSIFICATION OF ELEMENTS PERIODIC LAW

- The first 'periodic table' was proposed by 'Mendeleev'
- "The physical and chemical properties of the elements are periodic functions of their atomic weights"
- Mendeleev's periodic table is also known as short form of periodic table. (only 63 elements) while arranging the elements in the periodic table, he not only followed the increasing order of atomic weight but also considered their properties.
- Zero group elements were not known at the time of Mendeleev and later introduced by Ramsay and Rayleigh.
- Mendeleev has a foresight to leave some gap in the periodic table for 3-Elements and those are discovered later.



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**Contact: **96 76 57 8793****94 94 55 8793**

GET IT ON

Google Play

Download on the

App Store

Group	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII
Oxide Hydride	R ₂ O RH	RO RH ₂	RO RH ₃	R ₂ O ₃ RH ₄								
Periods	A	B	A	B	A	B						
1	H 1.008											
2	Li 6.939	Be 9.012		B 10.81	C 12.011	N 14.007	O 15.999	F 18.998				
3	Na 22.99	Mg 24.31		Al 26.98	Si 28.09	P 30.974	S 32.06	Cl 35.453				
4	First series: K 39.102	Ca 40.08		Sc 44.96	Ti 47.90	V 50.94	Cr 52.00	Mn 54.94	Fe 55.85	Cu 63.55	Zn 65.38	
Second series: Ga 69.72		Zn 65.38		Ge 72.64	As 74.92	Se 78.96	Br 79.904					
5	First series: Rb 85.47	Sr 87.62		Y 88.91	Zr 91.22	Nb 92.91	Mo 95.94	Tc 98	Ru 101.07	Rh 102.91	Pd 106.42	
Second series: Ag 107.87		Cd 112.40		In 114.82	Sn 118.69	Sb 121.75	Te 127.60	I 126.90				
6	First series: Cs 132.90	Ba 137.34		La 138.91	Hf 178.49	Ta 180.95	W 183.85		Os 190.2	Ir 192.22	Pt 195.08	

Mendeleev observed that elements with similar properties have almost same atomic weight - Fe(56), Co(59), Ni (59)

Atomic MassMendeleev-
proposed namePresent nameDiscovered by

44

Eka Boron

Scandium

Nilsson

70

Eka Aluminum

Gallium

De-Bios baudran

72

Eka Silicon

Germanium

Clemens Winkler

MERITS:

(1) The broadly left some gaps in discovered elements. It led to discovery of some new elements.

Eg: Ge, Si

(2) The atomic weights of Beryllium, Indium and Osmium by using corrected valence of elements.

Atomic weight = Equivalent weight × Valence

Demerits:

(1) Position of Hydrogen was not made clear.

(2) Isotopes are not included cause of periodicity is not known.

(3) Anomalous pairs of Mendeleev's periodic table were not explained

(a) Ar - K (b) Cobalt - Nickel (c) Te - I (d) Th - Pa

Anomalous pair = elements with higher atomic weight were placed before lower atomic weight.

MODERN PERIODIC TABLE:

→ Modern periodic table was proposed by Moseley based on atomic number.

$\sqrt{v} = a(z-b)$

z = atomic number

v = frequency of the x-rays

a, b are constants for an element

Follow **Chandan Logics** onDownload **Chandan Logics APP**Contact: **96 76 57 8793 / 94 94 55 8793**One **STOP** For **ALL Competitive EXAMS**

GET IT ON

Google Play

Download on the

App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

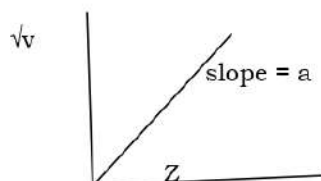
Contact: 96 76 57 8793

94 94 55 8793

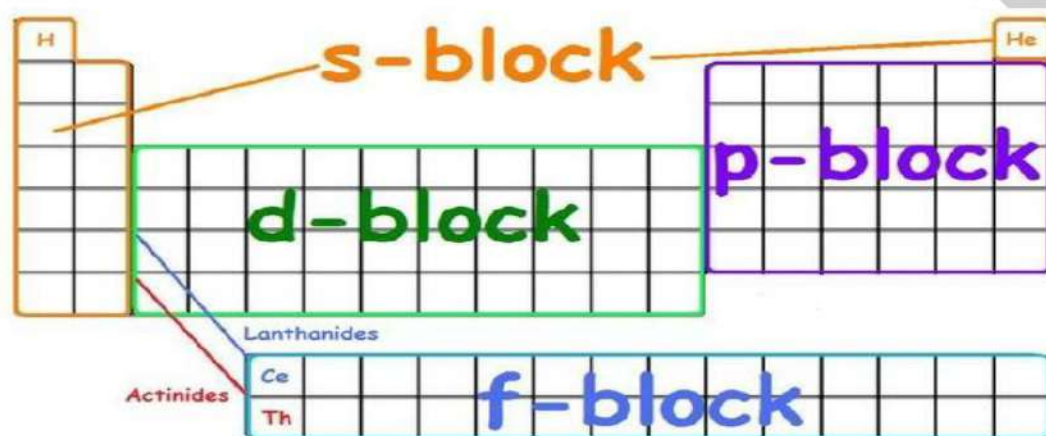
Get it on Google Play

Download on the App Store

As atomic number increases the frequency of characteristic x-rays increases. Atomic number has provided a better basis for the periodic arrangement of the elements.



Moseley discovered the atomic numbers from x-ray spectra of elements by bombarding the elements with cathode rays and the elements emitted respective x-rays of characteristic frequency.



Total number of groups (vertical columns) = 18

Total number of periods (Horizontal rows) = 7

- The shortest period - 1st period
- The longest periods - 6th & 7th periods.
- S-Block + P-Block = Representative elements

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL** Competitive **EXAMS**

Get it on Google Play

Download on the App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON Google Play

Download on the App Store

- d-Block = Transition elements
- f-Block = Inner transition elements
- Zn, Hg, Cd - are not transition elements due to the absence of partly filled d-orbital's both in atomic and in ionic states.
- Zinc, Mercury, Cadmium - are referred as non-typical transition elements (or) volatile metals.

1st Group Elements (Alkali Metals) - ns^1

		Atomic number
Hydrogen	H	1
Lithium	Li	3
Sodium	Na	11
Potassium	K	19
Rubidium	Rb	37
Cesium	CS	55
Francium	Fr	87

2nd Group (Alkaline Earth Elements - ns^2)

Be	Beryllium	4
Mg	Magnesium	12
Ca	Calcium	20
Sr	Strontium	38
Ba	Barium	56
Ra	Radium	88

3rd Group - Boron family - ns^2np^1

B	Boron	5
Al	Aluminium	13
Ga	Gallium	31
In	Indium	49
Tl	Thallium	81
Nh	Nihonium	113

IV- Group - Carbon family - ns^2np^2

C	Carbon	6
Si	Silicon	14
Ge	Germanium	32
Sn	Tin	50
Pb	Lead	82
Fl	Flerovium	114

V- Group - Pnictogens (Nitrogen family) - ns^2np^3

N	Nitrogen	7
P	Phosphorus	15
As	Arsenic	33
Sb	Antimony	51
Bi	Bismuth	83
Mc	Moscovium	115

Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google PlayDownload on the
App Store**VI- Group: Chalcogens (ore forming elements) - ns^2np^4**

O	Oxygen	8
S	Sulphur	16
Se	Selenium	34
Te	Tellurium	52
Po	Polonium	84
Lv	Livermorium	116

VII- Group: Halogens (Sea Salts) : ns^2np^5

F	Fluorine	9
Cl	Chlorine	17
Br	Bromine	35
I	Iodine	53
At	Astatine	85
TS	Tennessine	115

VIII Group: Inert gases / Noble gases / Zero group - ns^2np^6

He	Helium	2
Ne	Neon	10
Ar	Argon	18
Kr	Krypton	36
Xe	Xenon	54
Rn	Radon	86
Og	Oganesson	118

d - Block elements: $(n-1)d^{1-10} ns^{0-2}$

- Transition elements - $(n-1) d^{1-9} ns^{1-2}$
- These elements exhibit variable oxidation states.

3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8	9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B
21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38
39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.411
57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592
89-103	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [268]	110 Ds Darmstadtium [269]	111 Rg Roentgenium [272]	112 Cn Copernicium [277]

Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**GET IT ON
Google PlayDownload on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

Download on the

App Store

f - Block elements: Inner transition elements

- $(n-2) f^{1-14} (n-1) d^{0-1} ns^2$
- **Antepenultimate** = $(n-2)$
- **Pen ultimate shell** = $(n-1)$
- **Ultimate shell** = n

Eg: (1) What is the Anti, Penultimate shell of $6S^2$? $(n-1) = 5 d$ Penultimate shell $(n-2) = 4f$ Anti penultimate shell

57 La Lanthanum 138.905	58 Ce Cerium 140.116	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.243	61 Pm Promethium 144.913	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.925	66 Dy Dysprosium 162.500	67 Ho Holmium 164.930	68 Er Erbium 167.259	69 Tm Thulium 168.934	70 Yb Ytterbium 173.055	71 Lu Lutetium 174.967
89 Ac Actinium 227.028	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.080	99 Es Einsteinium [254]	100 Fm Fermium 257.095	101 Md Mendelevium 258.1	102 No Nobelium 259.101	103 Lr Lawrencium [262]

Lanthanides: - $4f$ orbital ----- $4f^{1-14} 5d^{0-1} 6s^2$

Atomic number 57 lanthanum (La) to 71 Lutecium (Lu)

Actinides: - $5f$ orbital - $5f^{1-14} 6d^{0-1} 7s^2$

Atomic number 89 Actium (AC) to 103 Lawrencium (Lr)

Periodic Properties:

The properties which are repeated after certain interval of atomic numbers are called periodic properties.

(1) **Atomic size:**

The distance between nucleus and the outermost electron of an atom is called atomic radius.

Units --- $\text{\AA}$ (angstrom)(2) **Electron Affinity (or) Electron Enthalpy:**

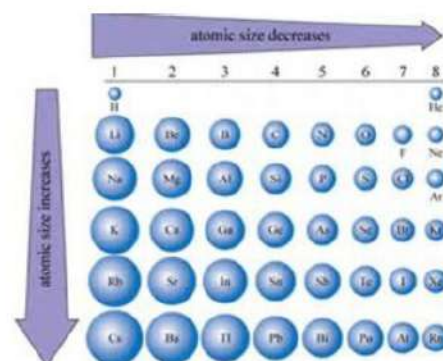
The amount of energy released when an electron is added to a neutral isolated gaseous atom of an element is called Electron Affinity.

(1) $Cl > F > Br > I > At$ (2) $S > Se > Te > O$ **Electro Negativity:** (E.N)

The tendency of an atom to attract the shared electron pair towards itself in a molecule is called electro negativity.

Electro Negativity can be measured with "Pauling Scale"

$$X_A - X_B = 0.208\sqrt{\Delta}$$

 X_A, X_B E.N of A & B Δ = Polarity of A-B bondFollow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL** Competitive **EXAMS**

GET IT ON

Google Play

Download on the

App Store

**Ionization Potential:** (I.P)

The amount of energy required when an electron is removed from a neutral Isolated gaseous atom of an element is called I.P.

**PERIODIC TRENDS**

Periodic property	Group (top to bottom)	Periods (Left to right)
Atomic size	increases	decrease
Electro positive nature	increases	decrease
Basic Nature of oxides	increases	decrease
Acidic Nature of oxides	decreases	increases
Non-Metallic Nature	decreases	increases
Electron Affinity	decreases	increases
Electro Negativity	decreases	increases
Ionization Potential	decreases	increases

CHEMICAL INDUSTRIES

1) SUGARCANE INDUSTRY:

Carbohydrates: Optically active Poly hydroxy aldehydes (or) Ketene derivatives.

It can be classified into 3 types.

(1) Monosaccharides (Simple sugars).

These cannot be further Hydrolyzed to simpler Molecules.

Eg: Erythrosine, threose, Glucose, Fructose, Ribose

(2) Oligosaccharides: It undergoes hydrolysis and yield 2 to 10 Monosaccharide units.**(3) Polysaccharides:** It undergoes hydrolysis and give more than 10 monosaccharide units.

Eg: Starch, Cellulose, glycogen, Dextrin, Gums.

Reducing sugar:

The saccharides that reduce felling's reagent, Tolle's reagent are called as reducing sugars.

It forms silver mirror with Tollens reagent and give red ppt with Fehling Solution.

Eg: Glucose, Fructose, trios, tetrose, pentose. All disaccharides are reducing sugars except sucrose.

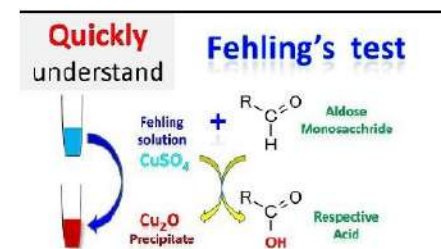
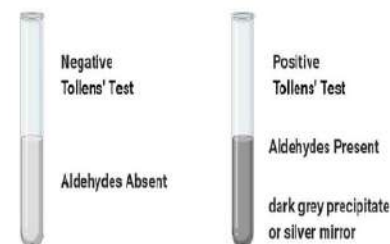
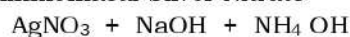
Non-Reducing sugar:

It cannot reduce Fehling's reagent and tollen's reagent.

Eg: Sucrose, starch, cellulose, glycogen dextrin.

Tollens's reagent:

Ammoniacal Silver Nitrate



Fehling's reagent:

Sodium Citrate + Sodium Carbonate + Copper Sulphate

Sugars can be classified into two types

- (1) **Natural Sugars**
- (2) **Artificial Sugars**

Natural Sugars:

Blood sugar / Grape Sugar / corn Sugar - Glucose

Cane Sugar / Table Sugar - Sucrose.

Milk Sugar - Lactose

Sweetest natural Sugar/ Honey sugar - Fructose



Artificial Sugars:

	<u>Name of Sugar</u>	<u>Sweeter than sugar</u>	
(a)	Aspartame	100 time	- Cool drinks & Soft drinks
(b)	Saccharin	550	- Sugar free tablets
(c)	Sucralose	600	
(d)	Alitame	2000	
(e)	Sucranoic acid	2 lakh time	

Sugar cane Industry

By Products:

- (a) Biogases - Rocket fuel
(b) Press mud - Fertilizer
(c) Molasses - 50% sucrose
(d) Carbon dioxide

Fermentation process: (Industrial preparation method of alcohol)

Molasses + yeast + water Ethanol + CO₂

Yeast Enzymes

- Invertase
- Zymase

Molasses (Sucrose) + H₂O **Invertase** **Glucose + Fructose**

Glucose / Fructose **Zymase** **Ethanol + CO₂**

96% alcohol - Rectified spirit / Industrial alcohol

- **100% alcohol is called as Absolute alcohol**
- **96% alcohol** **Azeotropic dist**

➤ **96% alcohol** **Azeotropic distillation** **100% alcohol**

$\xrightarrow{\Delta}$

Denatured alcohol → **100% alcohol + Pyridine** **drink** → **Permanent Blindness**

→ **100% alcohol + Methanol** → **Leads to debate**



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

➤ Fake alcohol / illicit alcohol = 100% alcohol + Methanol

The chemical can be used in Breathing Analyzers ---

Potassium dichromate ($K_2 Cr_2 O_7$)



Here ethanol can be converted into Ethanoic acid



Raw material	Type of liquor	Percentage of alcohol
Potato	Vodka	60%
Molasses	Rum	45 - 55%
Corn	Zin	40- 45%
Grapes	Brandy	40- 45%
Malt	Whisky	35 - 40%
Grapes	Wine	12 - 16%
Barley	Beer	3 - 6%

2) Food Preservators:

- Sodium Meta Biphosphate
- Sodium Bicarbonate
- Potassium Bicarbonate
- Citric acid
- Tartaric acid
- benzoic acid (or) sodium Benzoate

Different Food Preservation Methods



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

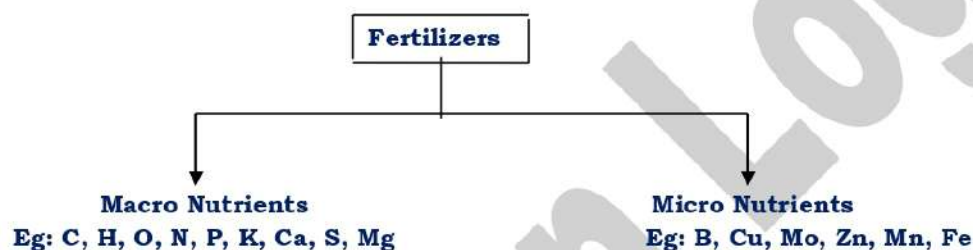
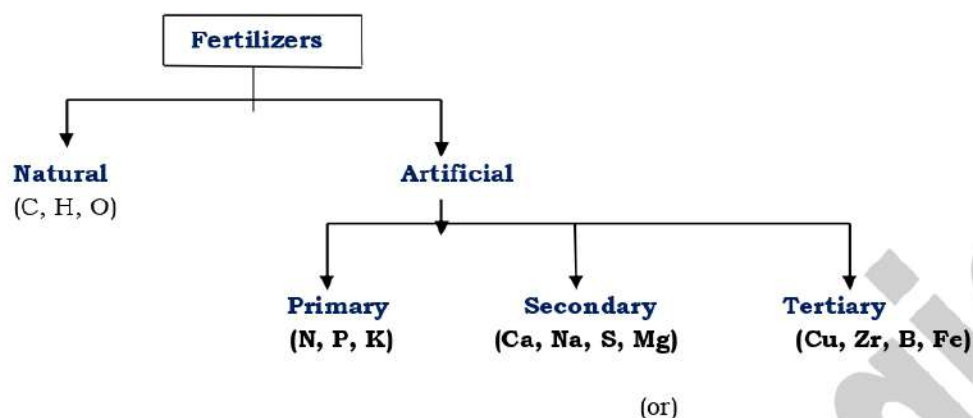
One **STOP** For **ALL** Competitive **EXAMS**

GET IT ON
Google Play

Download on the
App Store



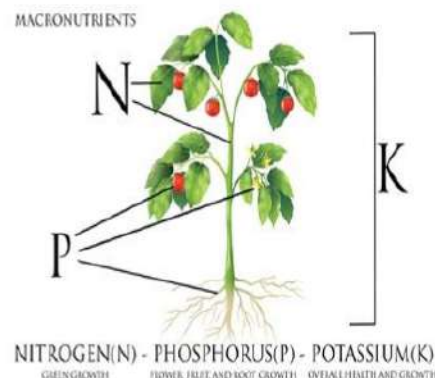
3) Fertilizer industry



'K' Suppliers : Potassium chloride (KCl)
Potassium sulphate K_2SO_4
Potassium Nitrate KNO_3

'P' suppliers: Superphosphate of Lime
Ammonium phosphate
Ammonium phosphate

Nitrogen suppliers : (1) Ammonia (N=82.35%)
(2) Urea (N= 46.66%)
(3) Nitrolim Co $(CN)_2$
(4) Mono ammonium phosphate (MAP)
(5) Di ammonium Phosphate (DAP)

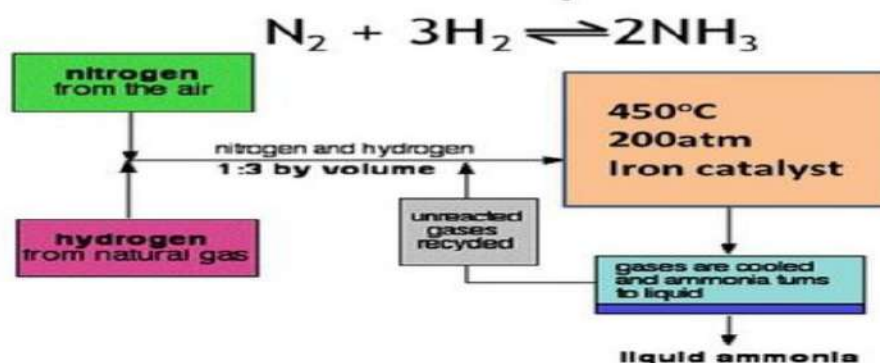


Haber is process (Industrial preparation method of Ammonia)





Making ammonia The Haber process



PERCENTAGE OF NITROGEN IN AMMONIA

$$N \% = \frac{\text{Nitrogen mass}}{\text{Total mass}} \times 100$$

$$\text{Ammonia (N\%)} = 14/17 \times 100 = 82.35\%$$

➤ The major nitrogen supplier is ammonia

ACIDS, BASES & SALTS

According to 'Robert Boyle' compounds can be classified into three types

- (a) Acids (b) Bases (c) Neutral

Acids: The word 'Acid' is derived from the Latin word 'acere' which means 'sour'

(1) Acids exhibit sour taste

Example:

Ant's sting	- Formic acid
Honey Bee's / Nettle leaves	- Formic acid (Methanolic acid)
Gastric Juice	- Hydrochloric acid
Urine	- Uric acid
Spinach/Tomato/watermelon	- Maleic acid
Curd	- Lactic acid
Amla	- Ascorbic acid (vit-c)
Grapes/Tamarind paste	- Tartaric acid
Muscle contraction	- Lactic acid
Snake poison	- Arsenic acid



(2) **Nonmetal oxides are Acidic in Nature**

Eg: CO_2 - Carbon dioxide
 NO_2 - Nitrogen dioxide
 SO_2 - Sulphur dioxide



Eg: $\text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{CO}_3$ (Carbonic acid)
 $\text{NO}_2 + \text{H}_2\text{O} \longrightarrow \text{HNO}_3$ (Nitric acid)

Note: HNO_3 - Aqua Fortis (Latin for strong water) being Highly corrosive, the solution was used in alchemy for dissolving silver and gold.

(3) **$\text{pH} < 7$**

Eg: Sulphuric acid (king of chemicals) - 0.1

(4) **Methyl orange Indicator** $\xrightarrow{\text{Acids}}$ **Red colour**
 (orange colour)

(5) **Phenolphthalein Indicator** $\xrightarrow{\text{Acids}}$ **Colour less**
 (Colour less)

(6) **Turmeric Powder (yellow)** $\xrightarrow{\text{Acids}}$ **Yellow Colour**
 (Natural Indicator)

(7) **China rose indicator** $\xrightarrow{\text{Acids}}$ **Dark Pink/Magenta**

(8) **Blue Litmus paper** $\xrightarrow{\text{Acids}}$ **Red Colour**

Red Litmus paper $\xrightarrow{\text{Acids}}$ **Red Colour**

➤ Raw material of Litmus paper - Lichen algae

(9) **Arrhenius theory: (Acid)**

An acid is the substance which provides Hydrogen ions in Aqueous Medium



Note:

The common element presents in all acids - 'H' spirit of salt - HCl

Demerits of Arrhenius Theory:

It can't explain the acidic nature of nonmetal oxides.

Eg: SO_2 , NO_2 , CO_2

Lewis Acids:

An acid is any molecule or ion that can accept a pair of electrons and forms a co-ordinate covalent bond.





Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

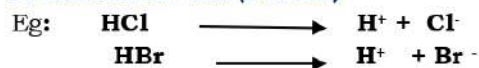
GET IT ON
Google Play

Download on the
App Store

Eg: H^+ , $AlCl_3$, BF_3

Bronsted - Lowry's Acid

It donates H^+ ion (Proton)



Conjugate Acids:

By the addition of H^+ ion, conjugate acid can be formed.

Eg:	Substance/ion	Conjugate Acid
	H_2O	H_3O^+ ($H_2O + H^+$)
	H_2S	H_3S^+ ($H_2S + H^+$)
	Cl^-	HCl ($H^+ + Cl^-$)
	OH^-	H_2O ($H^+ + OH^-$)
	O^{2-}	OH^- ($H^+ + O^{2-}$)
	S^{2-}	SH^- ($H^+ + S^{2-}$)



Importance of Acids

(1) The chemical which is used in etching of glass - Hydrofluoric acid (HF)

(2) Gold can be dissolved in

- (a) Mercury
- (b) Aqua Regia (Royal Liquid)
- (c) Hydrogen Cyanide

aqua Regia = $HCl + HNO_3$ (3:1)

➤ The name of mercuric Alloy is called Amalgam

(3) Acid rains Majorly caused by

- (a) Sulphur dioxide (SO_2) / Sulphuric acid
- (b) Nitrogen dioxide (NO_2) / Nitric acid
- (c) Carbon dioxide (CO_2) / Carbonic acid

P^H value < 5.6

The responsible one of Taj mahal colour change - Acid Rains

(4) Vinegar = 4% Acetic acid - Anti Microbial agent in Bakery Products (or) cheese.

(5) To restore old oil Lead Paints, the chemical used is Hydrogen peroxide.

(6) The chemical ingredients of soft drinks

- (a) Citric acid
- (b) Carbonic acid
- (c) Phosphoric acid

(7) The preservative of Green fruits - formic acid



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL** Competitive **EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

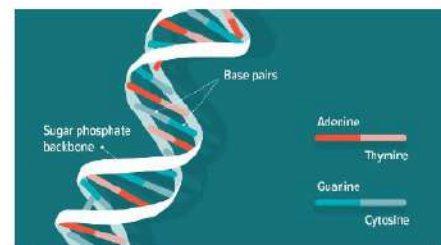
94 94 55 8793

GET IT ON Google Play

Download on the App Store

(8) **Butter has characteristic (or) pleasant smell due to the presence of Butyric acid.**

(9) Each cell in our body contains an acid the deoxy Ribonucleic acid (or) DNA. It controls every feature of the body such as our looks, colour of eyes, height etc.



(10) **Fat - Fatty acids - Longer chain of carboxylic acids.**

(11) **The strongest acid - Perchloric acid (HClO_4)**
Magic acid ($\text{HSO}_3\text{F} + 90\% \text{ SbF}_5$)

(12) **Lead Battery / car battery acid - Sulphuric acid**



(13) **Boric acid - Antiseptic agent**

(14) **Acids can be used as Food preservative**
Eg: Sorbic acid / benzoic acid - Pickles, Jams
Sodium Benzoate fruit juice
Iso Ascorbic acid/ citric acid - Soft drinks juice, wine



(15) **The nature of sweat - Acidic (Lactic acid)**

DIFFERENT CHEMICAL CAN BE USED TO REMOVE DIRT AND STAINS FROM CLOTHING

- | | | | |
|-----|---|-----|---|
| (a) | Rust marks | --- | Oxalic acid / citric acid |
| (b) | Tea, Coffee | -- | Citric acid / Borax powder
Chlorine water/ Sulphur dioxide |
| (c) | Ink | -- | Tartaric acid/ Hypo/ Chlorine water
sodium sulphate, sodium meta by sulphate |
| (d) | Blood | -- | Borax powder + Water
Ammonia + water
Starch + water |
| (e) | Oil/Grease / coal tar -
(Dry cleaning) | | Benzene /petrol/ kerosene/carbon tetrachloride
perchloro ethylene (or) Tetra chloro Ethylene |

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON Google Play

Download on the App Store

**Base:**

- (1)
- Bitter in taste and soapy in touch**

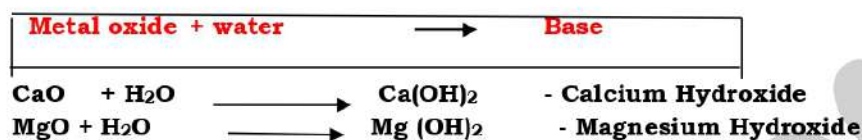
Eg: Caustic soda - NaOH - Sodium Hydroxide
Caustic potash - KOH - Potassium Hydroxide

These compounds can be used in Bio cremation (or) alkaline Hydrolysis.



- (2)
- Metal oxides are basic in nature**

Eg: Calcium oxide - CaO
Magnesium oxide - MgO



- (3)
- pH > 7**

Eg: Blood 7.32 - 7.45

- (4)
- Methyl orange indicator**
- Base →
- Yellow colour**
-
- (orange colour)

- (5)
- Phenolphthalein Indicator**
- Base →
- Pink colour**
-
- (colour less)

- (6)
- Red Litmus paper**
- Base →
- Blue colour**
-
- Blue Litmus paper**
- Base →
- Blue colour**

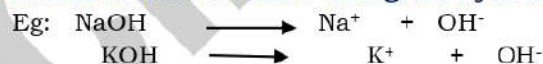
- (7)
- Turmeric powder**
- Base →
- Brownish red**
-
- (Yellow)

- (8)
- China Indicator**
- Base →
- Green**



- (9)
- Arrhenius Base:**

A Base is the substance that gives hydroxide ion in Aqueous medium



It can't explain the basic nature of metal oxide.

- (10)
- Lewis Base**
- : it donates a lone pair of electrons

Eg: NH₃ , OH⁻ , S²⁻ , Cl⁻ , Br⁻ , CN⁻

- (11)
- Bronsted - Lowry's Base:**

It accepts Hydrogen ion (H⁺) (or) proton

Eg: NH₃ , OH⁻ , S²⁻ , Cl⁻





Conjugate Bases : It can be formed by the removal of proton.

Eg:

Substance (or) ion	Conjugate Base
H_2O	OH^-
OH^-	O^{2-}
H_2S	SH^-
SH^-	S^{2-}
HNO_3	NO_3^-
H_2SO_4	HSO_4^-
HSO_4^-	SO_4^{2-}

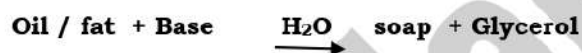
IMPORTANT OF BASES:

- (1) **An Antidote for Acidic poison - Antacids (bases).**

Eg: Milk of magnesia - $\text{Mg}(\text{OH})_2$ Magnesium Hydroxide.

Gelusil - Calcium Bicarbonate
Potassium bicarbonate
Sodium Bicarbonate
Aluminium Hydroxide

- (2) **Saponification: (Industrial preparation Method of Soap)**



- (3) **Aluminum Hydroxide** - Fire proof (or) water proof cloths

- (4) **The main ingredient in tooth paste - Calcium carbonate**

- (5) **NH_4OH (Ammonium Hydroxide) is used as - window cleaner**

- (6) **The best adsorbent of air pollutants (CO_2 , SO_2) - Potassium Hydroxide (KOH)**

- (7) **When an Ant bites, it injects the acidic liquid into the skin. It can be neutralized by rubbing moist baking soda / calamine solution (zinc carbonate).**

- (8) **Plants do not grow well, when the soil is either too acidic or too basic when it is too Acidic treated with base like Calcium oxide, calcium hydroxide. When it is too basic treated with organic matter.**

Salt : (Neutralization method)



NEUTRALIZATION ENERGY

- (1) **Strong Acid + Strong Base = Neutral Salt + 13.7 Kcal/Mol**
 (2) **Strong Acid + weak base = Acidic salt + 13.4 Kcal/ Mol**
 (3) **Weak Acid + Strong base = Basic salt + 13.4 Kcal/Mol**

Follow Chandan Logics on



Download Chandan Logics APP

Contact: 96 76 57 8793 / 94 94 55 8793

One STOP For ALL Competitive EXAMS





Chandan Logics

Chandan Logics

Aspire to Inspire

Download Chandan Logics APP

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

Download on the

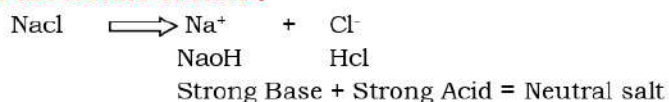
App Store

(4) Weak acid + weak base = cannot define

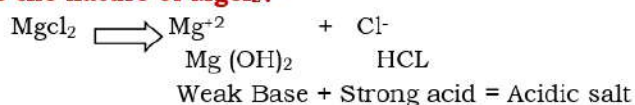
Strong Acids - H_2SO_4 , HClO_4 , HI , HBr , KCl , HNO_3

Strong Base: NaOH , KOH , CSOH , Ca(OH)_2 , Sr(OH)_2 , Ba(OH)_2

What is the Nature of NaCl ?



What is the nature of MgCl_2 ?



IMPORTANCE OF SALTS

- (1) The compound which can be used for blood clotting - Potash Alum
- (2) The fixing agent in photograph - Hypo $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$
- (3) Copper sulphate - $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ - Insecticide / Fungicide
- (4) Calcium chloride - Dehydrating agent
- (5) The compound frequently used in Leather Industries for the purification of water - potash alum.
- (6) The compound which is used in the manufacture of core components of the crackers - Potassium Nitrate (KNO_3)
- (7) Vitriol (Latin name of sulphate) 'Vitriol' comes from the Latin word vitriolum.
 - White Vitriol - $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ - Zinc sulphate
 - Green Vitriol $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ - Iron sulphate
 - Blue Vitriol $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ - Copper sulphate
 - Vitriol of syrup $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 - Oil of vitriol / spirit of vitriol - H_2SO_4
 - Sweet oil of vitriol --- Diethyl ether
 - Vitriol of mars ---- Iron (III) Sulphate

IONIC PRODUCT OF WATER (K_w)

The product of $[\text{H}^+]$ and $[\text{OH}^-]$ is known as Ionic Product of water.

$$K_w = [\text{H}^+] [\text{OH}^-]$$

@ 25°C $[\text{H}^+] = [\text{OH}^-] = 10^{-7}$ Mole ion / Lt

$$K_w = 10^{-7} \times 10^{-7} = 10^{-14}$$

→ K_w depends only on temperature

→ Temperature ↑ then K_w ↑

→ By the addition of acid or base ' K_w ' value can't be changed.

Follow Chandan Logics on



Download Chandan Logics APP

Contact: 96 76 57 8793 / 94 94 55 8793

One STOP For ALL Competitive EXAMS

GET IT ON

Google Play

Download on the

App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

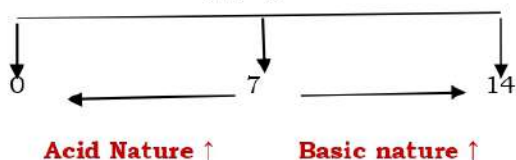
Download on the
App Store

pH Scale (Potentiality of Hydrogen = P^H)

The P^H value of solution is numerically equal to the logarithm of the inverse of the Hydrogen ion (H^+ concentration).

→ P^H scale was introduced by Sorensen

$$P^H = -\log [H^+]$$



$P^H < 7$ Acids

$P^H > 7$ Bases

$P^H = 7$ Neutral

$P^H = \text{constant}$ (Buffer)

<u>SUBSTANCE</u>	<u>P^H VALUE</u>
Sulphuric acid	0.1
Gastric juice	1.2
Citric acid	2.2
Soda water	2.4
Vinegar	3
Beer	4.3
Coffee	5
Egg white	5.6
Soft drinks	5.5 - 6.8
Milk	6.6
Rain water	7
Blood	7.32 - 7.45
Seawater	8
Baking soda	8.2
Milk of magnesia	10.6
Ammonia	11.4
Washing soda	12.8
Saliva (before Meals)	7.4
Saliva (after meals)	5.8

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

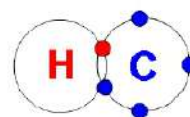
One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



ORGANIC CHEMISTRY



- It is the branch of chemistry which deals with study of Carbon compounds.
- The amount of carbon in the earth crust - 0.3%
- According to vital force theory organic compound can't be prepared in the lab artificially.
- The first organic compound is urea which was synthesized in the laboratory by Wohler

$$\text{NH}_4\text{CNO} \xrightarrow{\text{Tautomerism}} \text{NH}_2\text{CONH}_2$$

Ammonium cyanate

- The first organic compound which is synthesized from its constituent elements - Acetic acid (by Kolbe)
- **The King of elements -- Carbon**
- Carbon compounds are innumerable due to
 - Maximum catenation capacity of Carbon
 - Tetra valency of carbon
 - Isomerism in organic compounds of Carbon
 - Carbon has tendency of form multiple bonds

(a) **Catenation:** The self-linkage of atoms of same elements to produce long chain compounds is called catenation.

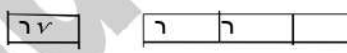
(1) -C-C-C-C-C-C-C-C

E.g. (2) C₆₀, P₄, S₈

Note: The element with highest catenation capacity is carbon = C₆₀

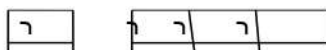
(b) **TETRA VALENCE OF CARBON:**

Carbon atomic number = 6 1s² 2s² 2p² Valence shell

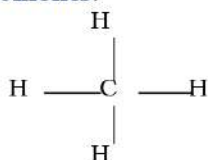


Excited state

1s² 2s¹ 2p³

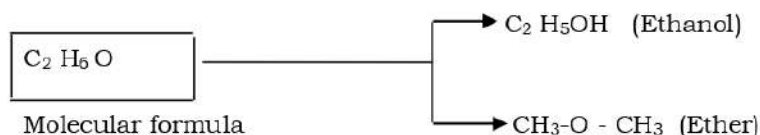


The number of unpaired electrons in the excited state of carbon is 4. So it forms four bonds (Tetra valence) with other elements.



(c) **ISOMERISM:**

The compound having same Molecular formula but differ in structural formula & properties are known as Isomerism.

(d) **MULTIPLE BOND FORMATION:**

Carbon has tendency to form single, double & triple bonds

- (1) $-C-C-$ ----- Saturated Hydro Carbons
- (2) $-C=C-$ ----- Unsaturated Hydro Carbons
- (3) $-C \equiv C-$ ----- Unsaturated Hydro Carbons

Allotropy: The same element exists with same chemical properties but different physical properties

(or)

The phenomenon of existence of an element in two or more physical forms having similar chemical properties but different physical properties is called allotropy.

E.g.

Oxygen

Phosphorus

Sulphur

O_2 (para magnet)

O_3 (dia magnet)

White phosphorus

Red Phosphorus

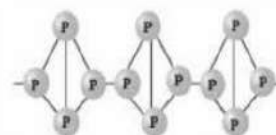
Rhombic Sulphur

Plastic Sulphur

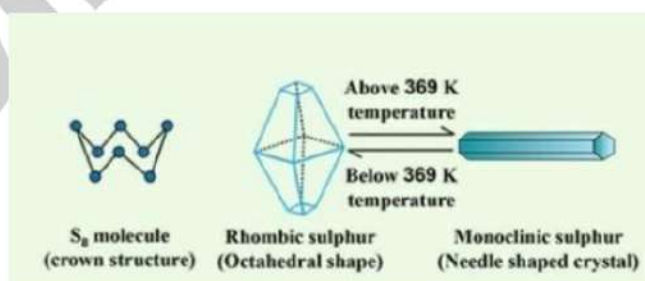
Mono clinic Sulphur

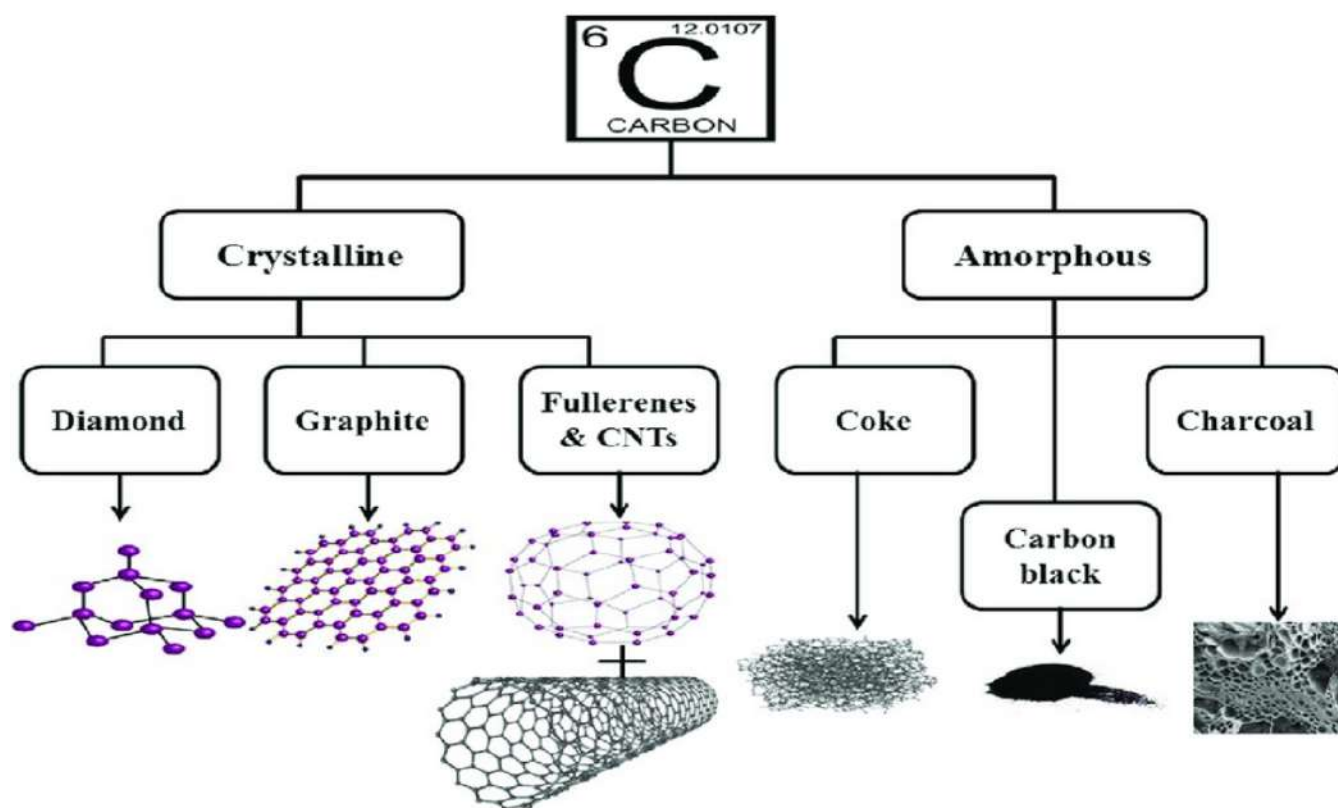


White phosphorus



Red phosphorus





DIAMOND

- (a) Hybridization - sp^3
 (b) Shape- Tetrahedral
 (c) Bond Length (c-c) $\Rightarrow 1.54 \text{ \AA}$ (1 angstrom = 10^{-10}m)
 (d) Bond angle $\Rightarrow 109^\circ 58'$

IMPORTANCE:

- It is an element
- Diamonds were first founded in India
- A deadly poisonous element
- The velocity of sound is maximum in diamonds only (12000 m/sec)
- It has Highest refractive index $\mu = 2.42$
- It is considered as one of the strongest material due to presence of Covalent Bonds.
- Diamonds are electrical insulators and extremely efficient thermal conductors.
- Most of the diamonds colour would be colorless. But some shows black, pink, blue, yellow etc.
- The thermal conductivity of diamonds at lower temperature (104 K) - 41000 watts per meter Kelvin
- Diamonds weight can be represented through carats
 - 1 carat = 200mg / 0.2 gm / 100 cents/ 100 parts
 - Eg: **Kohinoor Diamond = 105 carats**
 - Cullinane Diamond = 3030 carats**



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

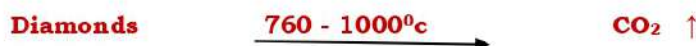
94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

Hopp Diamond = 445 carats

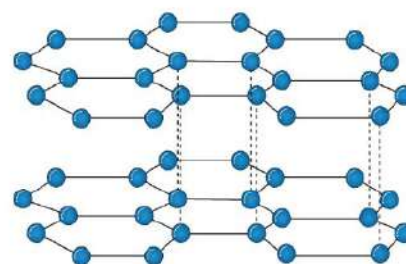
- Diamonds are used to make space probe windows, glass cutters, Gem cutlers etc.
- It is chemically inactive. But it forms CO_2 when it is heated with potassium dichromate & Sulphuric acid at 2000°C



- Artificial diamonds are made up with Silicon Carbide (SiC)

GRAPHITE

- Hybridization - SP^2
- c-c bond length $1.39\text{\AA}$
- Bond angle 120°
- Geometry = Hexagonal



Importance

- (1) It is thermally more stable allotrope of Carbon.
- (2) It acts as a good conductor of electricity due to the presence of free electrons (P_z orbital)
- (3) It is used as a dry Lubricant in Heavy Machines running at High Temperature, where oil cannot be used as a lubricant.
- (4) It is used as a moderator in nuclear reactors.
- (5) The percentage of Lead in Lead Pencils - Zero
- (6) The chemical ingredients of Lead Pencils - Graphite clay

- Aquadag is a mixture of Carbon powder + water
- It is used as electrically conductive coating on insulating surface and as a Lubricant.



GRAPHENE:

- (1) It is an element.
- (2) 2-dimensional form of carbon is considered as graphene.
- (3) The Lightest, thinnest and most flexible material
- (4) The strongest material than diamond
- (5) One of the Best Super conducting material.
- (6) The best material to prevent corrosion (or) rusting
- (7) The best material to convert salt water (or) sea water into drinking water.



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

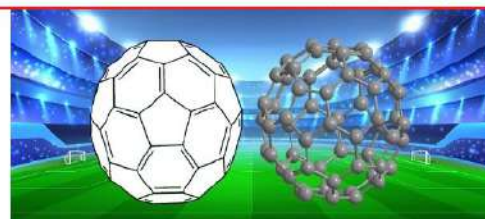
Download on the
App Store

BUCK MINISTER FULLERENE:

- Geometry - Foot Ball (or) soccer ball
- C-C Bond length $1.4\text{\AA}$
- It is an element
- It has pentagonal and hexagonal rings, resemble the balls used in football.
- C_{60} is also called Bucky Ball
- It can be used in the stealth technology of Indian missiles

Eg: ASTRA (Air to Air Missile)

This type of missiles are called as "Beyond visual of Radar and Air to Air Missile" (BVRAAM)



TYPES OF COAL:

Coal variety	Carbon percentage	Calorific value
Peat	57%	5700 kcal/kg
Lignite	60-70%	6500 - 7000 kcal/kg
Bituminous	78 - 90%	8000 - 85000 kcal / kg
Anthracite	90-98%	8700 kcal / kg

The oldest form of coal - Anthracite

- Anthracite has lowest volatile matter.

Calorie: The amount of Heat required to raise the temp of 1 gm of water through 1°C (15°C - 16°C)

Note: animal charcoal is used in sugar industry for decolorization of sugar because it absorbs Coloured material.

Carbon forms more than 1 crore compounds because of catenation and isomerism property.

Coconut charcoal is best adsorbent than Animal charcoal.

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

Download on the

App Store

HYDRO CARBONS

The derivatives of Hydrogen and carbon is called Hydro carbons.

E.g. Methane CH_4

Ethane C_2H_6

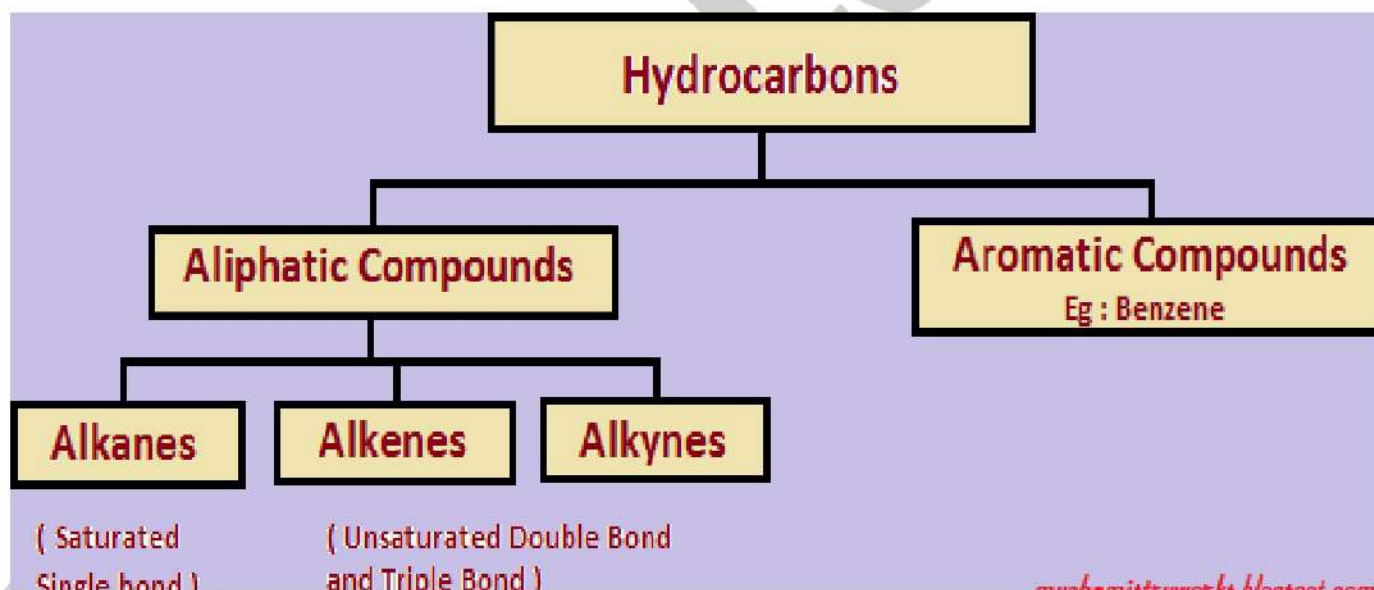
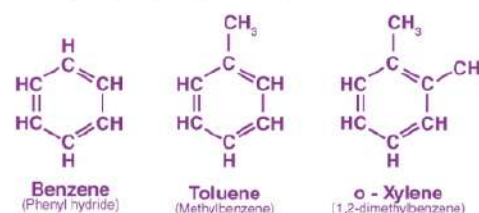
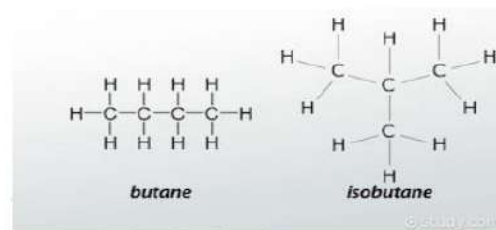
These are classified into two types

(1) Aliphatic Hydrocarbons (chain form):

Aliphatic hydrocarbons are hydrocarbons based on chains of C atoms.

(2) Aromatic Hydrocarbons (ring form):

Aromatic hydrocarbons, or arenes, are aromatic organic compounds containing solely carbon and hydrogen atoms. The configuration of six carbon atoms in aromatic compounds is called a "benzene ring", after the simple aromatic compound benzene, or a phenyl group when part of a larger compound.



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL** Competitive **EXAMS**

GET IT ON

Google Play

Download on the

App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

Download on the

App Store

Aliphatic Hydro carbons are classified into 3 types

<u>Alkenes</u>	<u>Alkenes</u>	<u>Alkynes</u>
(1) C-C Saturated	C = C Unsaturated	C = C unsaturated
(2) General formula C _n H _{2n+2} Eg: CH ₄	C _n H _{2n} C ₂ H ₄	C _n H _{2n-2} C ₂ H ₂
(3) Paraffin's	Olefins	-----
(4) Covalent Bond	Covalent Bond	Covalent bond
(5) Substitution reactions	Addition reactions	Addition reaction
(6) C ₁ - C ₅ - Gas C ₆ - C ₁₀ - Liquids >10 - Solids	1-3 → gas 4-13 → liquid >13 → solid	1-3 - gas 4-11 -liquid > 11 - solids



→ The C-C bond length is shortest in -----?

(a) Ethane (b) Ethylene (c) Benzene (d) Acetylene

FUNCTIONAL GROUPS

Formula

R - OH

R - CHO

R - COR

R - O - R¹

R - COOH

R - NH₂R - COOR¹

R - CN

R - Nc

Name of the group

Alcohol

Aldehyde

Ketene

ether

Carboxylic acid

Amine

Ester

Cyanide

Isocyanide

Examples.

Methanol

Ethanol

Propane

diethyl ether

acetic acid

Methyl amine

Methyl ethanoate

Methyl Cyanide

Methyl Isocyanide

Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON

Google Play

Download on the

App Store

CARBOHYDRATES

OPTICALLY ACTIVE POLY HYDROXYL, ALDEHYDES (OR) KETONES (OR) THE COMPOUNDS WHICH YIELD CARBOHYDRATES ON HYDROLYSIS.

Classification of carbohydrates into 3 types

(1) MONO SACCHARIDES (SIMPLE SUGARS) :

These can't be further hydrolyzed to simpler molecules.

Eg: Erythrose, Threose, Glucose, Fructose, ribose
20 monosaccharides occur in nature

(2) OLIGOSACCHARIDES:

These undergoes hydrolysis and yield 2 to 10 monosaccharide units.

Eg: Sugar Sucrose, Raffinose, Maltose, lactose, Galactose

(3) POLYSACCHARIDES:

These undergo hydrolysis and give more than 10 monosaccharide units.

Eg: starch, cellulose, Glycogen, dextrin, Gum, General formula $(C_6H_{10}O_5)_n$

The carbohydrates are also classified as either reducing or non-reducing sugar.

- 1) Reducing sugars
- 2) Non-reducing sugars

1)REDUCING SUGARS

The saccharides that reduce Fehling's reagent Tollen's reagent are called as reducing sugars.

They form silver mirror with Tollens reagent and give red ppt with Fehling's solution.



FEHLING'S REAGENT



Eg: Glucose, fructose, triose, tetrose, Pentonores.

All disaccharides are reducing sugar's except sucrose

NON REDUCING SUGARS:

The saccharide which does not reduce Fehling's reagent and Tollen's reagent are called non-reducing sugars.

Eg: Sucrose, starch, cellulose, glycogen, dextrin

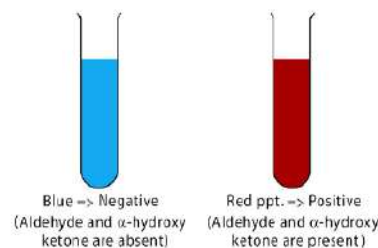
Based on sweetness sugars can be classified into two types

(1) NATURAL SUGARS: [Glucose, Fructose (Honey bees)]

- Blood sugar - Glucose
- Fruit sugar - Fructose
- Grape sugar - Glucose
- Corn sugar - Glucose



Fehling's Test



**(2) ARTIFICIAL SUGARS:**

Saccharin - It is about 550 times sweeter than it is not biodegradable in the body and does not give any calorific value of food. It is excreted as such in urine.

It is primarily used as a sweetening agent by diabetic patients

Aspartame	-	100	- Cool drinks & soft drinks
Saccharin	-	550	Sugar free tablets
Sucralose	-	600	
Alitame	-	2000	
Sucranoic acid	-	2 lakh	

SUGAR INDUSTRY:

Sugar cane $\longrightarrow$ Cane juice + Bagasse (Pulp)

Cane juice + $\text{Ca}(\text{OH})_2$ $\longrightarrow$ Defecation + Press mud (reduce Acidic nature)

Cane Juice + CO_2 $\longrightarrow$ CaCO_3 + Cane juice $\longrightarrow$ press mud (Carbonization)

Cane juice + $\text{CaO} + \text{SO}_2$ $\longrightarrow$ Sulfatation $\longrightarrow$ Press mud

Cane Juice $\longrightarrow$ Centrifugal forcible machines $\longrightarrow$ sugar + Molasses (50% sucrose)

Main by products

- (1) Biogases
- (2) Press Mud (Fertilizer)
- (3) Molasis
- (4) CO_2

FERMENTATION PROCESS: (Industrial preparation method of alcohol)

Molasis + H_2O + yeast $\longrightarrow$ $(\text{NH}_4)\text{SO}_4, (\text{NH}_4)_3\text{PO}_4$

Yeast secretes two types of Enzyme. Those are

- 1) Invertase Enzyme
- 2) Zymase Enzyme

(1) Sucrose + H_2O $\xrightarrow{\text{invertase}}$ Glucose + Fructose



(2) Glucose $\xrightarrow{\text{Zymase}}$ $2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$

$\rightarrow$ The Byproduct of sugar cane industry is CO_2

96% Alcohol is main ingredient of liquors ethyl alcohol is commonly known as alcohol.
HENCE IT IS CALLED AS SPIRIT OF WINE.



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

DE NATURATED SPIRIT :

In order to prevent consumption of pure alcohol

- methyl alcohol or pyridine is added to it consumption of Denaturated alcohol causes permanent Blindness.
- Bootleg alcohol / illicit alcohol / fake alcohol spirit + CH_3OH
- Chloral hydrate for foam and diazepam for the narcotic effect are added to adulterate toddy.

The chemical can be used in breathing analyzer to find the alcoholic percentage is "Potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$)"



RAW MATERIAL	TYPE OF LIQUOR	PERCENTAGE OF ALCOHOL
Potato	Vodka	60%
Molassis	Rum	45-55%
Corn	Zin	40-45%
Grapes	Brandy	40-45%
Malt	Whisky	35-40%
Grapes	Wine	12-16%
Barley	Beer	3-6%

- The responsible chemical for the characteristic smell of sea is 'dimethyl sulphide (CH_3)₂
- The common name of bullet proof material is Kevlar
- The chemical name of Kevlar is poly-para phenyl terepathalamide
- Kevlar is 5 times stronger than steel and used to make bullet proof vests.



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

Download on the

App Store

CLOSED CHAIN COMPOUNDS

In these compounds first & last carbons are attached with each other.

Eg: cyclo propane Δ

(A) HOMO CYCLIC COMPOUNDS

These are the compounds in which the complete ring is formed by carbon atoms only.

These are two types.

(i) ALI CYCLIC COMPOUNDS:

These are the compounds having the properties like aliphatic compounds. These may be saturated (or) unsaturated like aliphatic compounds.

Δ Cyclo propane ; Δ Cyclo propane ; - Cyclo butane

(ii) AROMATIC COMPOUNDS:

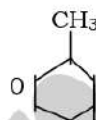
These compounds consist of at least one benzene ring

→ Aroma (Greek word, Aroma means sweet smell)

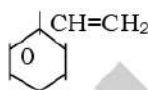
Eg: Benzene



Toluene



Styrene



HETERO CYCLIC COMPOUNDS: These are cycle compounds having ring of rings built up of more than one kind of atoms.

Furan



Thiophene



Pyrrole



GROUPS:

Atom or a group of atoms which passes any charge on it (or) any 'free valency' is called as groups.

ALKYL GROUP:

When a hydrogen is removed from saturated hydro carbons then alkyl group is formed.

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON

Google Play

Download on the

App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

Download on the

App Store

It is represented by 'R' & its General formula is C_nH_{2n+1}

Eg: $CH_4 \xrightarrow{-H} CH_3 - CH_2$ Ethyl

$CH_3 - CH_3 \xrightarrow{-H} CH_3 - CH_2$ Ethyl

(a) C_3H_7 has following two Isomers

(1) normal propyl $CH_3 - CH_2 - CH_2 -$

(2) Iso propyl $CH_3 - \underset{\begin{array}{c} | \\ CH_3 \end{array}}{CH} -$

(b) C_4H_9 has following four Isomers

(1) n-butyl $CH_3 - CH_2 - CH_2 - CH_2 -$

(2) Iso butyl $CH_3 - \underset{\begin{array}{c} | \\ CH_3 \end{array}}{CH} - CH_2 -$

(3) Secondary butyl $CH_3 - CH_2 - \underset{\begin{array}{c} | \\ CH_3 \end{array}}{CH} -$

(4) Tertiary butyl $\begin{array}{c} CH_3 \\ | \\ CH_3 - C - \\ | \\ CH_3 \end{array}$

ALKENYL GROUP:

$CH_2 = CH -$ Vinyl

$CH_2 = CH - CH_2 -$ Allyl

$CH_3 - CH = CH -$ Propenyl (1-propenyl)

$CH_3 - C_1 = CH_2$ Iso propenyl (1-methyl-1-ethyl)

ALKYNYL GROUP

$CH \equiv C -$ Ethynyl

$CH \equiv C - CH_2 -$ propargyl (2-propynyl)

$CH_3 - C \equiv C -$ (1-Propynyl) Propynyl

Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON

Google Play

Download on the

App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

Download on the

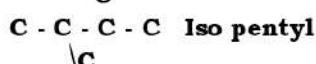
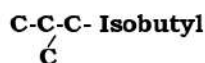
App Store

NORMAL GROUPS:

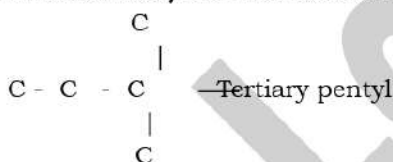
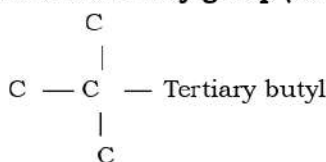
it is represented by 'n' (normal group - straight chain)

n - butyl $\text{C} - \text{C} - \text{C} - \text{C} -$ n - propyl $\text{C} - \text{C} - \text{C} -$ **ISO GROUP :**

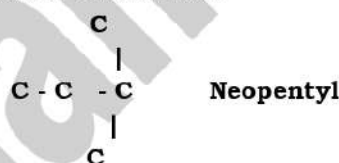
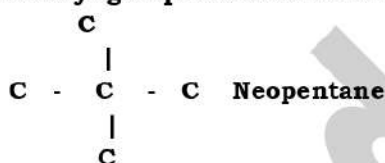
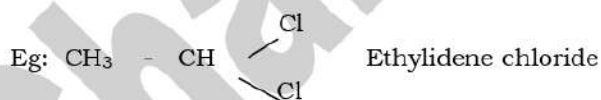
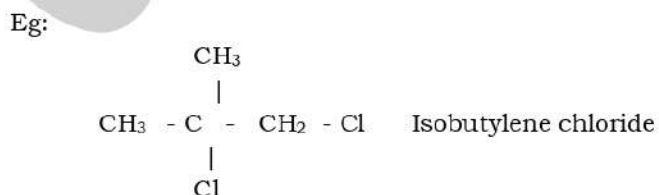
Two methyl groups are attached to the same carbon atom.

**TERTIARY GROUP:**

When three alkyl group (similar or dissimilar) are attached to the same Carbon atom.

**NEO GROUP:**

When four alkyl groups are attached to the same carbon atom

**COMMON - NAMING OF DIHALIDES**(1) **GEM DIHALIDES** - Two same halogen atoms are attached to the same Carbon(2) **VICINAL DIHALIDES** - Two same halogens are attached to adjacent carbon.Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON

Google Play

Download on the

App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

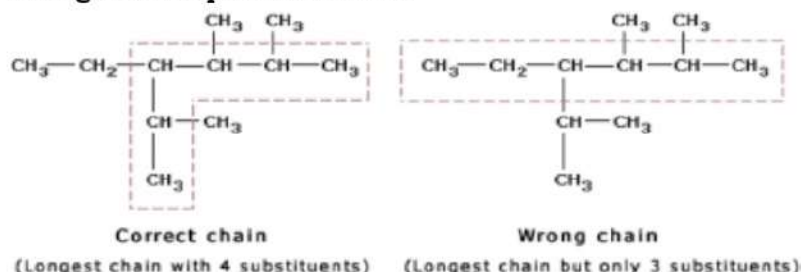
Download on the

App Store

IUPAC SYSTEM OR GENEVA SYSTEM

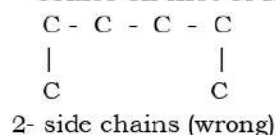
IUPAC RULES

(1) **LONGEST CHAIN RULE:** According to it in any organic compound the longest carbon chain is selected through all the possible chains.

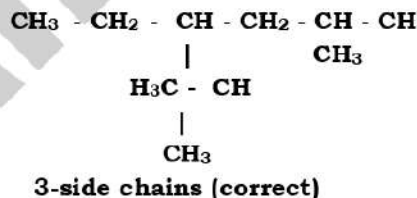
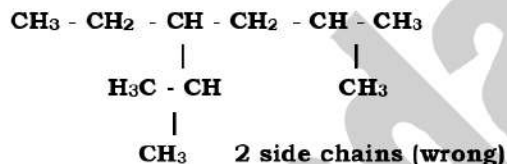


(2) Side Chains :

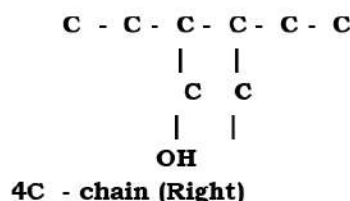
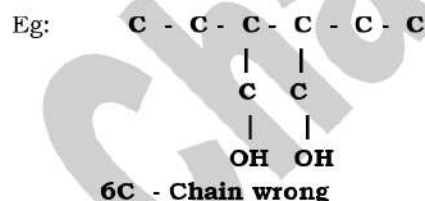
Those alkyl groups which are not included in the longest chain are called side chains. These never comes on first or last carbon (except in the naming of alkyl group)



(3) If there are the possibilities of longest chain more than one, the chain which possess more than one side chain is selected.



(4) The carbon which is attached with functions group have been included in the chain selected irrespective of the length of the chain.

Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL** Competitive **EXAMS**

GET IT ON

Google Play

Download on the

App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

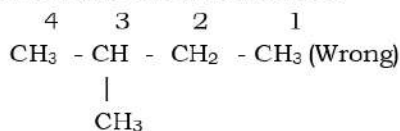
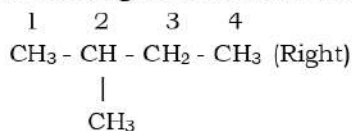
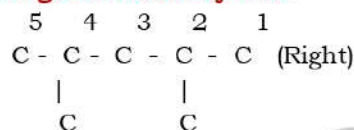
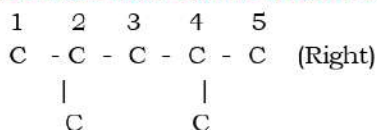
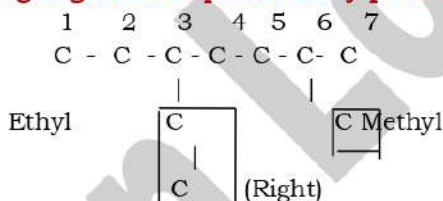
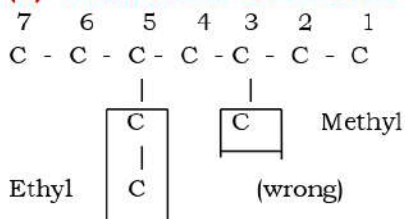
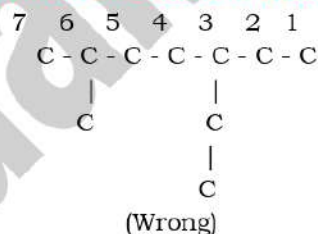
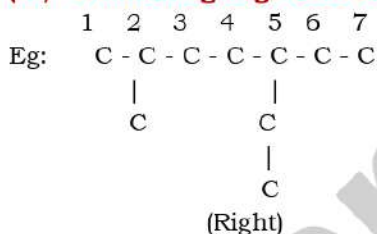
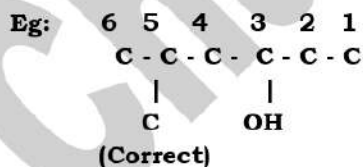
Google Play

Download on the

App Store

(5) FOR SATURATED HYDRO CARBONS:

Numbering is done from the end from which side chain is nearest.

**(a) If side chains are at same carbon from both sides, following rules are adopted.****(i) for same side chains numbering may be given from any end.****(ii) If side chains are different numbering is given in alphabetically preference order.****(iii) Numbering is given from the end from which side chain is nearest.****(D) FOR COMPOUNDS HAVING FUNCTIONAL GROUPS****(i) Numbering is given from the end from which functional group is closest.**Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL** Competitive **EXAMS**

GET IT ON

Google Play

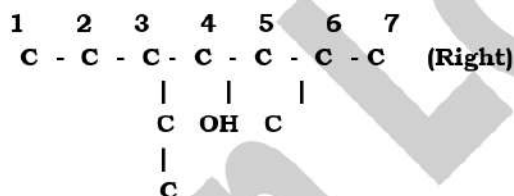
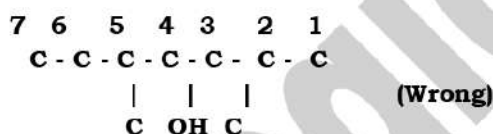
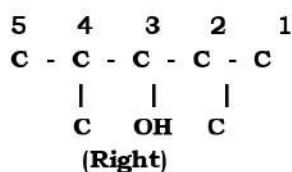
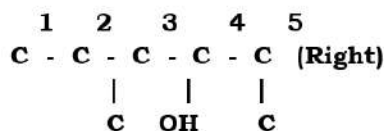
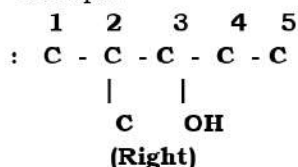
Download on the

App Store



- (ii) If the functional group is situated at same carbon from both sides, follow the rules of saturated hydro carbons.

Example:



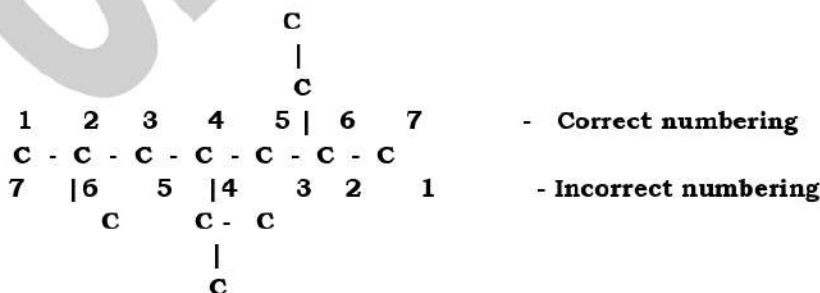
IMPORTANT RULES

(a) Initially write the number of that carbon on which the side chain is attached then write it in alkyl group.

- (b) If there are more than one side chains (same) initially write the numbers altogether using symbol (,) in between them, after last number use the symbol hyphen (-) write the numbers in increasing order.



(c) IUPAC name is written according to English alphabets (strictly)





Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

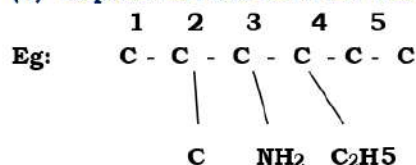
Download on the

App Store

2-methyl - 4 - Isopropyl - 5 Ethyl heptane (wrong)

5- Ethyl - 4 - Isopropyl - 2 - Methyl heptane (right)

(d) Capital work is used for first substitute



3 - Amino - 4 ethyl - 2 methyl

hexane

S.No.	Hydrocarbon Bond	Name
1.	(-)	Alkane
2.	(=)	Alkene
3.	(≡)	Alkyne
4.	2(=)	Alkadiene
5.	2 (≡)	Alkadiyne
6.	(=) + (≡)	Alken- yne

THE PRIORITY ORDER OF FUNCTIONAL GROUPS

S.No.	Name	Functional group formula	Prefix	Suffix
1.	Carboxylic	- C - OH	Carboxy/ Carboxylic	-oic acid
2.	Sulphonic acid	- SO ₃ H	Sulpho	-sulphonic acid
3.	Anhydride	- C - O - C	--	oic anhydride
4.	Ester	- C - O - R	Alkoxy Carbonyl	- oate
5.	Acyl halide	- C - X	Halo carbonyl / formyl	Oyl halide
6.	Amide	- C - NH ₂	Amido / Carbamoyl	-amide
7.	Cyanide	- C ≡ N	Cyano	Nitrile
8.	Iso cyanide	- N ≡ C	Iso cyano	Iso nitrile
9.	Alde hyde	- C - H	Formyl / oxo	al
10.	Ketone	- C -	Oxo / keto	-one
11.	Alcohol	- OH	Hydroxy	-ol
12.	Thioalcohol	- SH	Mercapto	Thiol
13.	Amine	- NH ₂	Amino	-
14.	Halo	- X	Halo	-
15.	Ether	- O -	Alkoxy	-
16.	Double bond	C = C	-	Ene
17.	Triple bond	C ≡ C	-	Yne

Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

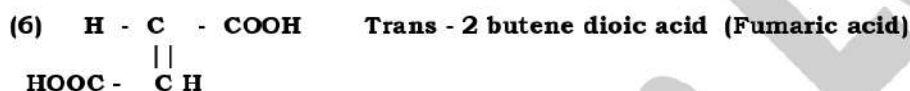
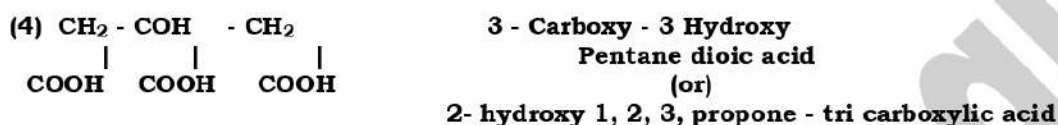
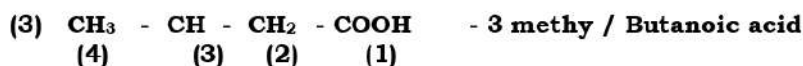
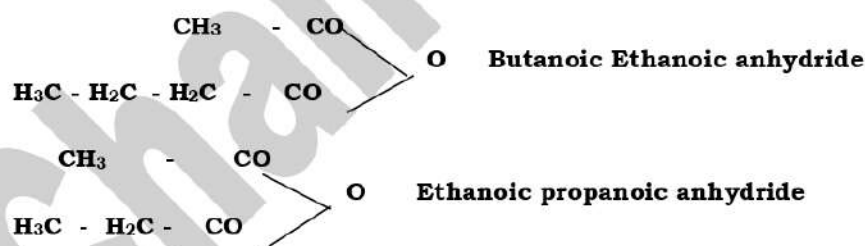
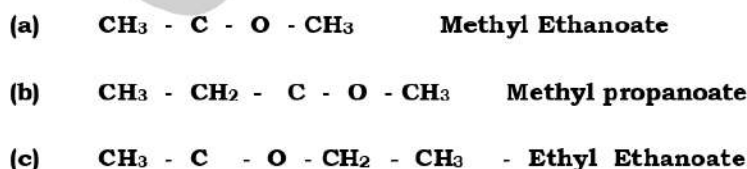
One **STOP** For **ALL Competitive EXAMS**

GET IT ON

Google Play

Download on the

App Store

Carboxylic group(2) SULPHONIC ACID:(3) ANHYDRIDE(4) ESTER

**(5) ACYL HALIDE**

- (a) $\text{CH}_3 - \text{C}(\text{O}) - \text{Cl}$ - Ethanoyl chloride
- (b) $\text{H} - \text{C}(\text{O}) - \text{Cl}$ - Methanoyl chloride

(6) AMIDE

- (a) $\text{CH}_3 - \text{C}(\text{O}) - \text{NH}_2$ Ethan amide
- (b) $\text{H} - \text{C}(\text{O}) - \text{NH}_2$ Meth amide
- (c) $\text{CH}_3 - \text{CH}(\text{CH}_3) - \text{C}(\text{O}) - \text{NH}_2$ 2 - Methyl propanamide
- (d) $\text{NH}_2 - \text{C}(\text{O}) - \text{C}(\text{O}) - \text{NH}_2$ Ethane diamide

(7) CYANIDE

- (a) $\text{CH}_3 - \text{CN}$ - Ethane Nitrile
- (b) $\text{CH}_3 - \text{CH}_2 - \text{CN}$ - Propane Nitrile
- (c) $\text{CH}_3 - \text{CH}(\text{CH}_3) - \text{CH}_2 - \text{CH}_3$ - 2 methyl Butane Nitrile
- d) $\text{CH}_2(\text{CN}) - \text{CH}_2(\text{CN}) - \text{CH}_2(\text{CN})$ 1, 2, 3 tricyano propane

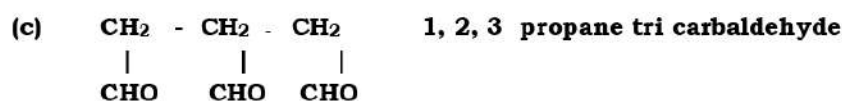
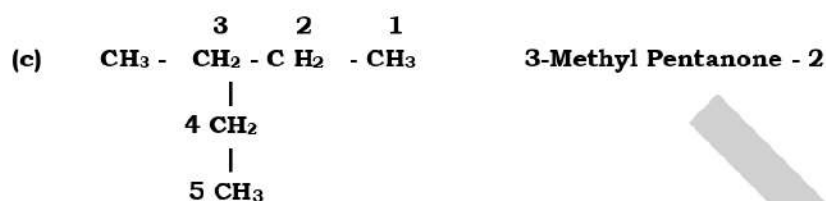
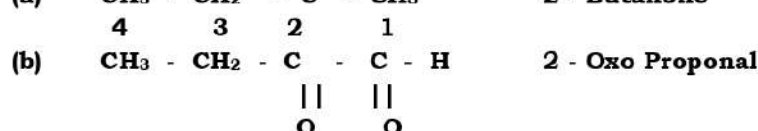
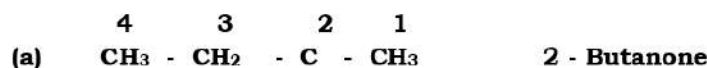
(8) ISOCYANIDE

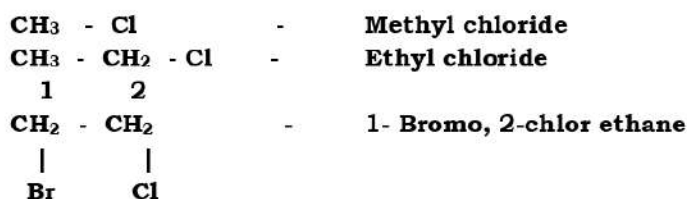
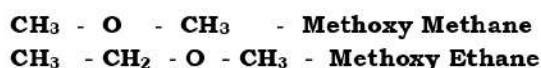
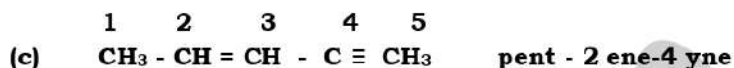
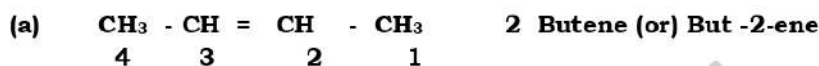
- (a) $\text{CH}_3 - \text{N} \equiv \text{C}$ Methane Iso nitrile (old)
Carbyl amino methane (new)
- (b) $\text{CH}_3 - \text{CH}_2 - \text{NC}$ - Carbyl amino Ethane.

(9) ALDEHYDE

- (a) $\text{CH}_3 - \text{CHO}$ Ethanal
- (b) $\text{CH}_3 - \text{CH}(\text{Br}) - \text{CHO}$ - 2- Bromo Propanal



**(10) KETONE:****(11) ALCOHOL****(12) THIOALCOHOL****(13) AMINE**

**(14) HALO****(15) ETHER****(16) DOUBLE BOND & TRIPLE BOND****BITS**

(1) How many secondary carbon atoms does methyl cyclopropane have?

- (A) None (B) One (C) Two (D) Three

(2) When one H atom is removed from alkyne we get?

- (A) Alkyl indene (B) Alkenyl (C) Alkynyl (D) Alkylidene

(3) Butyl shows ----- types of alkyl groups

- (A) 2 (B) 4 (C) 5 (D) 3

(4) The compound $\text{Cl}-\text{CH}_2-\text{C}-\text{OCH}_3$ will be classified



- (A) Halide (B) Ketone (C) Ester (D) Ether

(5) The structure of Acrolein is

- (A) $\text{CH}_3-\text{CH}_2\text{CHO}$ (B) $\text{CH}_3-\text{CH}=\text{CH}-\text{CHO}$ (C) $\text{CH}_2=\text{CH}-\text{CHO}$ (D) None

(6) Formula of vinyl methanoate is

- (A) $\text{CH}_3 - \text{C} - \text{O} - \text{CH} = \text{CH}_2$ (B) $\text{CH}_3-\text{CH}_2-\text{C}-\text{O}-\text{CH}_3$
 (C) $\text{CH}_2 = \text{CH} - \text{O} - \text{C} - \text{H}$ (D) $\text{H}-\text{C}-\text{O}-\text{CH}_2-\text{CH}=\text{CH}_2$





Chandan Logics

Chandan Logics

Aspire to Inspire

Download Chandan Logics APP

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON

Google Play

Download on the

App Store

- (7) **IUPAC name of the compound $\text{CH}_2 = \text{CH} - \text{CH}(\text{CH}_3)_2$ is**
 (A) 1,1 - dimethyl - 2 propane (B) 3- methyl -1 - butane
 (C) 2 - Vinyl propane (D) 1-isopropyl ethylene
- (8) **The IUPAC name of the compound having structure**

$$\begin{array}{ccccccc} \text{C}_2\text{H}_5 & - & \text{C} & - & \text{CH} & - & \text{CH}_3 \\ & & | & & | & & \\ & & \text{CH}_2 & & \text{CH}_3 & & \end{array}$$
 (A) 3-methyl-2ethyl butene-1 (B) 2-ethyl-3methyl butene-1
 (C) 3-ethyl-3 methyl butene-1 (D) ethyl isopropyl ethane
- (9) **Choose the correct IUPAC name for $\text{CH}_3 - \text{CH} - \text{CHO}$**

$$\begin{array}{c} | \\ \text{CH}_2\text{CH}_3 \end{array}$$
 (A) buton-2 aldehyde (B) 2-methyl butanal
 (C) 3-methyl isobutyroldehyde (D) 2-ethyl Proponal
- (10) **The IUPAC name of ether, $\text{CH}_2\text{OC}_2\text{H}_5$ is**
 (A) Ethoxy methane (B) Methoxy ethane
 (C) Methyl ethyl ether (D) Ethyl methyl ether
- (11) **The name of $\text{CH} = \text{CH}$ is**

$$\begin{array}{cc} | & | \\ \text{CHO} & \text{NH}_2 \end{array}$$
 (A) 1-amino prop-2-enal (B) 3-amino prop-2-enal
 (C) 1-amino - 2 formyl ethene (D) 3-amino-1-oxoprop-2ene
- (12) **The IUPAC name for the formula $\text{CH}_3 - \text{C} - \text{C} - \text{COOH}$**

$$\begin{array}{ccc} & \text{CH}_3 & \text{H} \\ & | & | \end{array}$$
 (A) 2- methyl - 2-butenic acid (B) 3-methyl - 3-butenic acid
 (C) 3-methyl - 2 butanoic acid (D) 2-methyl -3-butenic acid
- (13) **I.U.P.A.C name of**  **is**
 (A) 4,5 - dimethyl Oct- 4 -end (B) 3,4- dimethyl Oct-5-ene
 (C) 4,5 - dimethyl Oct -5-ene (D) none.
- (14) **The I.U.P.A.C. name of $(\text{CH}_3)_3\text{C} - \text{CH} = \text{CH}_2$ is**
 (A) 2,2 - dimethyl but-3-ene (B) 2,2 - dimethyl pent-4-ene
 (C) 3,3 - dimethyl but -1-ene (D) hex - 1 - ene
- (15) **The correct name f 3,3 - dimethyl propanamide is**
 (A) 2-methyl butanamide (B) 3-methyl butanamide
 (C) iso-propyl ethanamide (D) iso propyl acetamide

Follow Chandan Logics on



Download Chandan Logics APP

Contact: 96 76 57 8793 / 94 94 55 8793

One STOP For ALL Competitive EXAMS

GET IT ON

Google Play

Download on the

App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download Chandan Logics APP

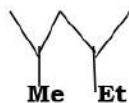
Contact: 96 76 57 8793

94 94 55 8793

GET IT ON Google Play

Download on the App Store

(16)



Me Et its I.U.P.A.C name is;

- (A) 2,4 - dimethyl hexane (B) 2-ethyl - 4-methyl pentane
(C) 2,4 - dimethyl Pentane (D) None of these

(17) The I.U.P.A.C. name of acetyl acetone is

- (A) 2,5 - Pentane dione (B) 2,4 - Pentane dione
(C) (D)

(18) The IUPAC name of $(\text{CH}_2 = \text{CH} - \text{CH}) \text{CH}_2 = \text{CH} - \text{C} \equiv \text{CH}$ is

- (A) 1-Butenyne - 3 (B) 1-Butynene-3
(C) Vinyl acetylene (D) all of them

(19) I.U.P.A.C. name will be $\text{CH}_2 - \text{CH} - \text{CH}_2$ 

- (A) 1,2,3 - Tricyano propane (B) Propane trinitrile - 1,2,3
(C) 1,2,3 - cyano propene (D) 3-cyano pentane - 1,3-dinitrile

(20) The simplest alkene containing three primary carbon atoms is

- (A) iso butylene (B) P - butylene (C) α - butylene (D) iso pentene

(21) Which of the following is crotonic acid

- (A) $\text{CH}_2 = \text{CH} - \text{COOH}$ (B) $\text{C}_6\text{H}_5 - \text{CH} = \text{CH} - \text{COOH}$
(C) $\text{CH}_3 - \text{CH} = \text{CH} - \text{COOH}$ (D) $\text{CH} - \text{COOH}$
|
 $\text{CH} - \text{COOH}$

(22) $\text{CH}_3 - \text{CH} - \text{C} - \text{CH} - \text{OCH}_2 \text{CH}_3$ the IUPAC name is

- (A) 2-ethoxy - 4 - methoxy pentan - 3 - one
(B) 2-methoxy - 4 - ethoxy Pentan - 3 - one
(C) 2-ethoxy - 4 - methoxy Pentan - 3 - one
(D) None of these

(23) IUPAC name of  is

- (A) But - 2 - ene-2, 3-diol (B) Pent - 2 - ene- 2, 3- diol
(C) 2-methyl but-2-ene-2, 3 - diol (D) Pent - 3 - ene-3, 4-diol

Follow Chandan Logics on



Download Chandan Logics APP

Contact: 96 76 57 8793 / 94 94 55 8793

One STOP For ALL Competitive EXAMS

GET IT ON Google Play

Download on the App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download Chandan Logics APP

Contact: 96 76 57 8793

94 94 55 8793

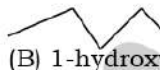
GET IT ON Google Play

Download on the App Store

- (24) IUPAC name of $\text{OHC}-\text{CH}=\text{CH}-\text{CH}-\text{CH}=\text{CH}_2$ is
- $$\begin{array}{c} | \\ \text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3 \end{array}$$
- (A) 4-butyl-2,5-hexadien-1-al (B) 5-vinyloct-3-en-1-al
(C) 5-vinyloct-5-en-8-al (D) 3-butyl-1,4-hexadien-6-al

- (25) The correct IUPAC name of $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$
- $$\begin{array}{cc} | & | \\ \text{C}_2\text{H}_5 & \text{C}_2\text{H}_5 \end{array}$$
- (A) 1,2-diethyl butane (B) 2-ethyl 3-methyl pentane
(C) 3,4-dimethyl-3-hexane (D) none is correct

- (26) The IUPAC name of $\text{CH}_3-\text{CH}_2-\text{CH}-\text{COOC}_2\text{H}_5$ is
- $$\begin{array}{c} | \\ \text{CH}_3 \end{array}$$
- (A) 2-ethyl-ethyl acetate (B) ethyl-3-methyl butanoate
(C) ethyl 2-methyl butanoate (D) 2-methyl butanoic acid ethylester

- (27) IUPAC name of the compound  OH is
- (A) But-2-en-1-al (B) 1-hydroxy but-1-ene

- (28) The IUPAC name of $\text{CH}_3\text{CH}_2-\text{N}-\text{CH}_2\text{CH}_3$ is
- $$\begin{array}{c} | \\ \text{CH}_3 \end{array}$$
- (A) N-Methyl-N-ethylethyl amine (B) diethyl methyl amine
(C) N-ethyl-N-methyl ethan amine (D) methyl diethyl amine

- (29) The IUPAC name of $\text{N}\equiv\text{C}-\text{CH}_2-\text{CH}_2-\text{OH}$ is
- (A) 1-hydroxy ethane nitrile (B) 3-hydroxy propane nitrile
(C) 2-hydroxy ethyl cyanide (D) dihydroxy-3-phenyl propanoic acid

- (30) The IUPAC name of $\text{C}_6\text{H}_5\text{CH}=\text{CH}-\text{COOH}$ is
- (A) Cinnamic acid (B) 1-phenyl-2-Carboxy ethene
(C) 3-phenyl prop-2-enoic acid (D) dihydroxy-3-phenyl propanoic acid

ANSWERS

- | | | | | | |
|--------|--------|--------|--------|--------|--------|
| (1) C | (2) C | (3) B | (4) C | (5) C | (6) 6 |
| (7) B | (8) B | (9) B | (10) B | (11) B | (12) C |
| (13) A | (14) C | (15) B | (16) A | (17) B | (18) A |
| (19) A | (20) A | (21) C | (22) A | (23) B | (24) A |
| (25) C | (26) C | (27) D | (28) C | (29) B | (30) C |

Follow Chandan Logics on



Download Chandan Logics APP

Contact: 96 76 57 8793 / 94 94 55 8793

One STOP For ALL Competitive EXAMS

GET IT ON Google Play

Download on the App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

ATOMIC STRUCTURE

ATOM: it is the smallest indivisible particle of an element that can take part in a chemical reaction

Atom is coined from **GREEK** word "**Atomio**" which means "**indivisible**" proposed by **Democritus**

ELEMENT: An element is a simple substance which can't be further divided into two or more substances.

Elements are divided as metals and non-metals the term element was first proposed by Robert Boyle.

DALTON'S ATOMIC THEORY:

The matter is made of small indivisible particles called atoms, which can take part in chemical reactions.

- The atoms of the same element are identical in size mass and in other properties in all respects.
- Atoms of different elements differ from each other in other their properties and masses
- Atoms of different elements can combine in simple ratio to form compounds.
- Atoms can be neither created nor destroyed.

MERITS:

Dalton's theory is able to explain law of conservation of mass, law of constant composition and law of multiple proportions.

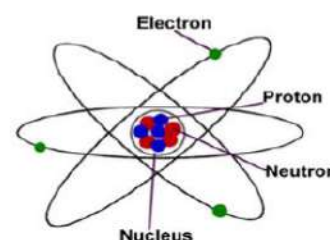
DRAWBACKS OF DALTON'S THEORY:

- It failed to explain the internal structure of atom.
- It could not explain how atoms of different elements differ from one another and combine with one another
- It failed to explain the experiments like when glass or ebonite rubbed with silk or fur generate electricity.

SUBATOMIC PARTICLES:

Atom consists of smaller particles like electron, proton, neutron, neutrino, antineutrino, positron, antiproton, pions and mesons etc.

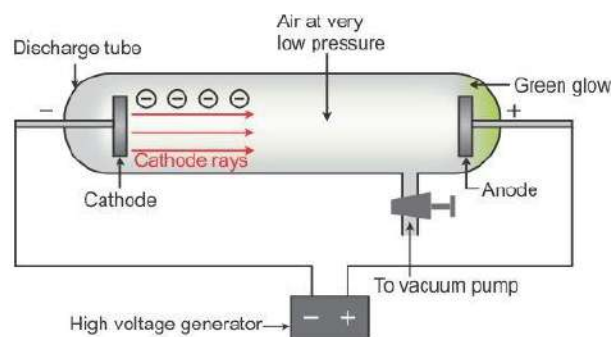
Electrons, Protons & neutrons are considered as fundamental particles of atom.



ELECTRON - DISCOVERY : (CATHODE RAYS)

It was discovered by J.J. Thomson.

- The name electron was proposed by Stoney
- The discharge tube is a sealed tube, made of glass containing two thin metal plates called electrodes.
- During the discharge tube experimental "Crookes" observed that rays were found to pass from negatively charged plate (cathode) to positively charged plate (anode)
- The electrical discharge through the gases could be observed only at very low pressure (0.01 mm of Hg) and at very high voltages (910000V).



Production of cathode rays
Cathode Rays Discharge Tube

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

→ the rays emitted from cathode are called cathode - rays, cathode rays are stream of electrons.

PROPERTIES OF CATHODE RAYS:

The cathode rays move from cathode to anode, these rays are invisible but their behavior can be observed with the help of fluorescent or phosphorescent materials, which glow when hit by them.

- These rays travel in straight lines in the absence of electrical and magnetic field.
- In the presence of electric field they are deflected towards anode and in magnetic field towards south pole.
- Cathode rays contain (-)ve charged particles called electrons.
- The nature of cathode rays is found to be independent of nature of the Cathode material and nature of the gas in the tube.
- Hence electrons are considered as constituent particles of all atoms.

CHARGE TO MASS RATION OF ELECTRON E/M

It is known as specific charge

- e/m increases then greater is the deflection when electric & magnetic field is applied.

Electron charge = -1.67×10^{-19} Coulombs: **Mass** = 9.0×10^{-28} gm (or)

0.00055 amu (Or) **Mass** = $1/1837$ of hydrogen

DISCOVERY OF PROTON : Discovered by Gold Stein

(Anode rays / +ve rays / canal rays)

- The name proton is proposed by Rutherford

PROPERTIES:

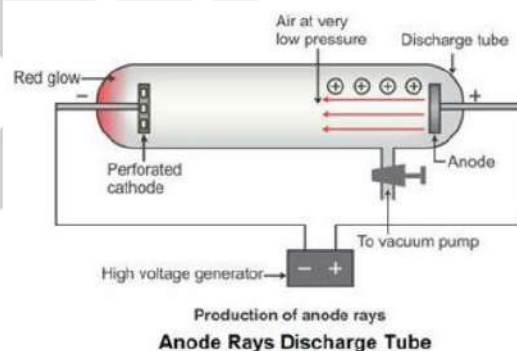
Anode rays travel in straight line and these are material particles.

e/m values less than electron

e/m value of anode rays depends upon nature of the gas it is maximum in Hydrogen medium

$m = 1.67 \times 10^{-24}$ gm

charge = 1.602×10^{19} coulombs



NEUTRON:

Neutrons are discovered by James Chadwick

- They have no charge but passes mass almost equal to proton
- | | |
|----------|-----------|
| electron | (-ve) |
| proton | (+ve) |
| Neutron | (neutral) |

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Name	Symbol	Charge	Relative Mass (amu)	Actual Mass (kg)
proton	p^+	+1	1	1.67×10^{-27}
neutron	n^0	0	1	1.67×10^{-27}
electron	e^-	-1	1/1840	9.11×10^{-31}

ATOMIC MODELS

J J THOMSON ATOMIC MODEL:

According to J J Thomson atom is like water melon, also called plum pudding raisin pudding model of atom.

- The (+ve) charge is distributed like fibrous material material and electron are embedded like seeds.

An atom is in spherical shape with radius approximately 10^{-10}m in which the (+) ve charge is uniformly distributed.

It can't explain electrical neutrality of the atom.

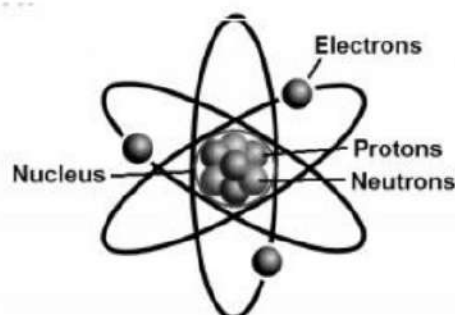
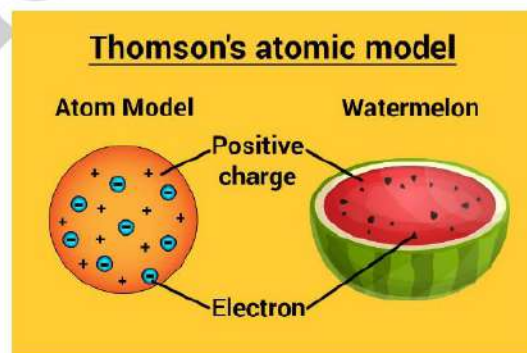
RUTHERFORD ATOMIC MODEL : (Planetary model) or (Nuclear Model)

radius of nucleus = 10^{-15}m / Fermi
radius of atom = 10^{-8}cm or 10^{-10}m (or) $1\text{\AA}$

Note: The smallest unit of distance - Fermi
Largest unit of distance - Parsec (3.26 light year)

DEMERITS / DEFECTS :

- Accordingly, to classical electromagnetic theory, the revolving electron should loss energy continuously and travel in a spiral path finally it must fall into the nucleus. But it does not happen.
- It was failed to explain stability of atom
- It can't explain the electronic structure of atom and energies of electrons
 $1 \text{ light year} = 9.4 \times 10^{12} \text{ km}$



Rutherford's Atomic Model



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

- The atomic spectrum should be continuously band spectrum due to continuous loss of energy but it alien spectrum.

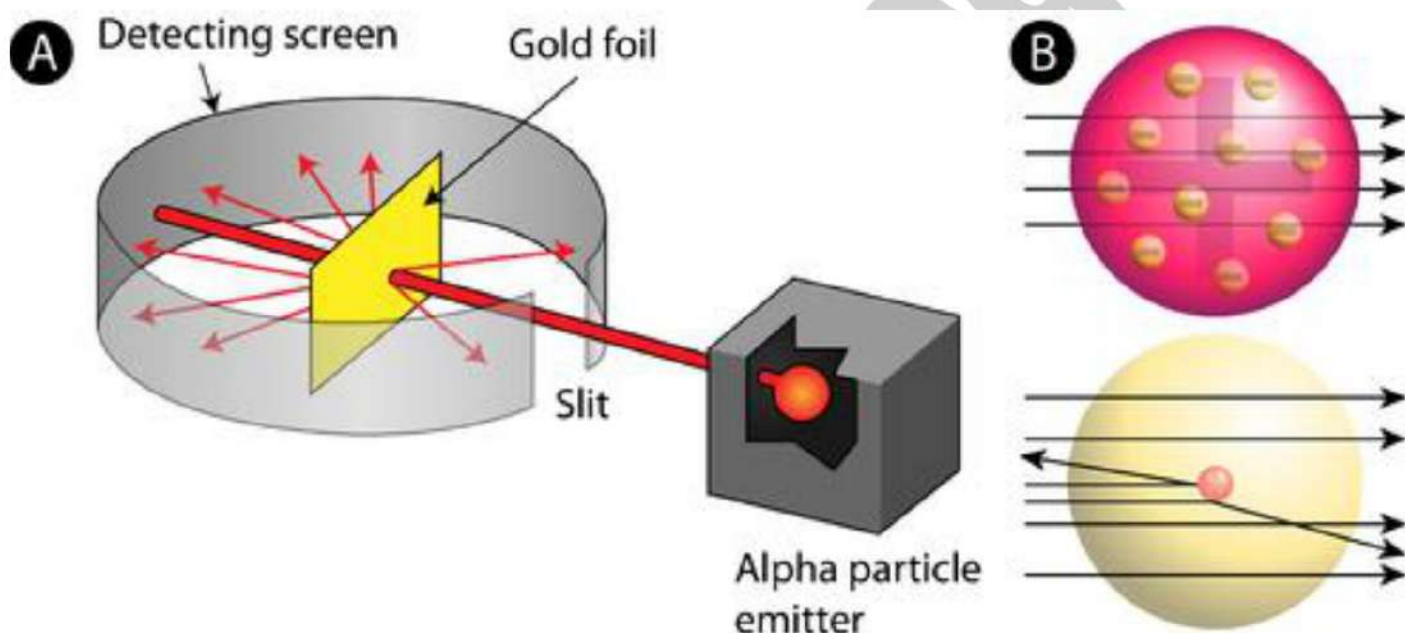
ATOMIC NUCLEUS:

When a beam of α -particles are passed through a thin gold foil which is enclosed with fluorescent ZnS screens capable of producing scintillations.

OBSERVATIONS:

Most of the α particles pass through the foil with any deflection which indicates that most of the atom is empty.

A small fraction of α - particles were deflected from its original path which indicates that the (+ve) is at the Centre of the atom and due to repulsive forces x deflections takes place.



A very few α -particles bounced back which indicates that the whole mass is concentrated at the Centre of atom. This heaviest part is name as "nucleus".

ATOMIC NUMBER (Z)

The no. of electrons are protons present in an atom of an element is called its Atomic number .

Moseley gives the relation b/w atomic number z and frequency of the characteristic x-rays of the element by the equation.

$$\sqrt{\nu} = a (z-b)$$

a, b are constants and depends on nature of the element

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**Contact: **96 76 57 8793****94 94 55 8793**GET IT ON
Google PlayDownload on the
App Store

mass number (A) = no of neutrons + no of protons

$$A = n + z$$

$$\text{no. of neutrons} = A - Z$$

ATOMIC WEIGHT:

One twelfth the mass of an atom of $^{12}_6\text{C}$ is called as atomic mass unit (a.m.u). It is also known as Dalton (Da)

H = 1	P = 31	Fe = 56
He = 4	S = 32	Co = 59
Li = 6.9	Cl = 35.5	Ni = 58.7
Be = 9	Ar = 40	Cu = 63.5
B = 11	K = 39	Zn = 65.3
C = 12	Ca = 40	
N = 14	Sc = 45	
O = 16	Ti = 48	
F = 19	V = 51	
Ne = 20	Cr = 52	
Na = 23	Mn = 55	
Mg = 24		
Al = 27		
Si = 28		

Calculate no. of Protons, Neutron & Electrons?

(1) $^{37}_{17}\text{Cl}$

(2) $^{14}_7\text{N}$

ISOTOPES:

The nuclides of same element having same atomic number but different mass numbers are called isotopes.

Isotopes show similar chemical properties but different physical and radioactive properties.

Eg: $^1_1\text{H}^1$ $^1_1\text{H}^2$ $^1_1\text{H}^3$

$^{35}_{17}\text{Cl}$ $^{37}_{17}\text{Cl}$; $^{235}_{92}\text{U}$ $^{238}_{92}\text{U}$

Follow **Chandan Logics** onDownload **Chandan Logics APP**Contact: **96 76 57 8793 / 94 94 55 8793**One **STOP** For **ALL Competitive EXAMS**GET IT ON
Google PlayDownload on the
App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

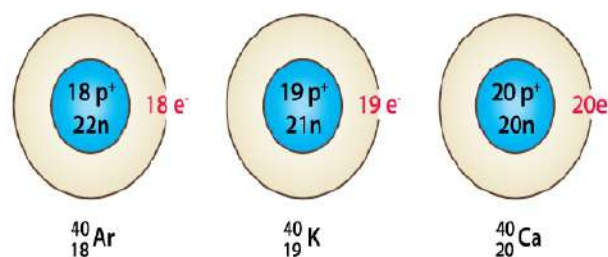
GET IT ON
Google Play

Download on the
App Store

ISOBARS

The nuclides of different elements having same mass number but different atomic numbers are called isobars.

It shows similar physical properties but different chemical properties.



ISOTONES :

The nuclides of different elements with different atomic number and mass number but having same no. of neutrons (A-Z) are called isotones.

(or)

Same no neutrons but different in protons electrons.

Eg: $_{14}\text{Si}^{30}$ $_{15}\text{P}^{31}$ $_{16}\text{S}^{32}$ (2) $_{6}\text{C}^{14}$, $_{8}\text{O}^{16}$

ISODIAPHERES :

The nuclides having same isotopic number (A-2Z) are called isodiapheres. They possess same difference of neutrons and protons (n-p)

$$_{9}\text{F}^{19} \quad A-2Z (9\text{F}^{19}) = 19 - 2 \times 9 = 1$$

$$_{11}\text{Na}^{23} \quad A-2Z (11\text{Na}^{23}) = 23 - 2 \times 11 = 1$$

ISOSTERS :

The molecules with same atomicity and same no. of electrons are called Isosters.

Eg: CO_2 , N_2O
 C_6H_6 , $\text{B}_3\text{N}_3\text{H}_6$

Atomicity: total no. of atoms in molecules.

ISO ELECTRONIC SPECIES:

The molecules or ions with same no. of electrons are called iso electronic species.

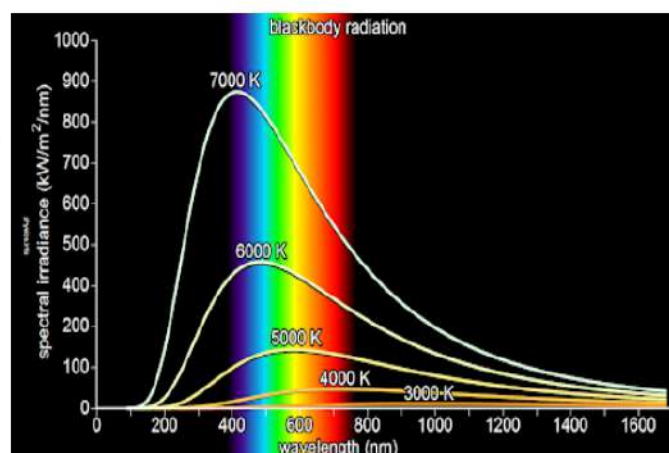
Eg: N^{3-} , O^{2-} , Ne , Na^+ , Mg^{+2} , Al^{+3}

BLACK BODY RADIATION:

The ideal body which emits and absorbs radiations of all frequencies is called a blackbody.

The radiation of blackbody is called black body radiation.

A hollow sphere coated inside with platinum black which has a small hole in its wall can act as a near black body.



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

QUANTUM THEORY: - (1900)

The phenomenon of the black body radiation is given by max plank.

→ The absorption or emission of radiation is mainly due to the vibrations of charged particles.

The radiation is associated with definite amount of energy.

→ The radiation energy is always in discontinuous wave form. It is absorbed or emitted discontinuously in the form small packets of energy called quantum.

$$E = h \nu$$

$$(or) E = hc / \lambda$$

E = Energy in ev

λ = wave length

Where E - energy of radiation = 6.625×10^{-27} erg Sec
h - plank's constant = 6.625×10^{-34} joul sec.

energy is emitted or absorbed in simple integral multiple of quantum, but not fractional values.

EINSTEIN - APPROACH: -

Accordingly, to Einstein the radiant energy is absorbed or emitted in the form of small particle is called photon. The radiation is propagated in the form of photons.

→ The energy of a photon is directly proportional to frequency of the radiation.

COMPTON EFFECT:

The increase in wavelength or ↓ in energy of the x-ray after scattering from an object is called Compton effect.

PHOTO ELECTRIC EFFECT:

The phenomenon of ejection of electrons from the surface of metal when light of suitable frequency strikes is called as photo electric effect.

This effect is readily exhibited by Alkali metal like K, Rb, Cs

A part of the energy of photon is used to escape the electron from the attractive forces and the remaining energy is used in increasing the K.E. of electron.

$$E = W + K.E$$

$$h \nu = h \nu_0 + KE$$

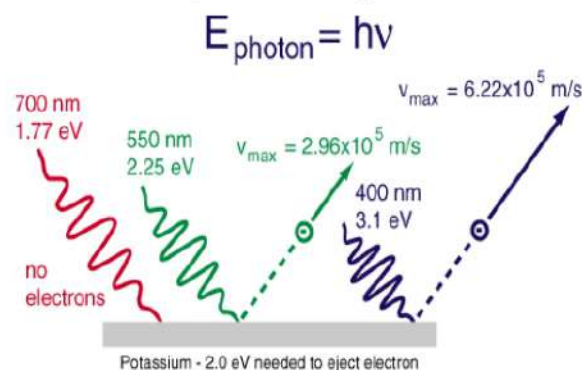
$$KE = h \nu - h \nu_0 = hc / \lambda - hc / \lambda_0$$

BOHR'S ATOMIC MODEL: (1913)

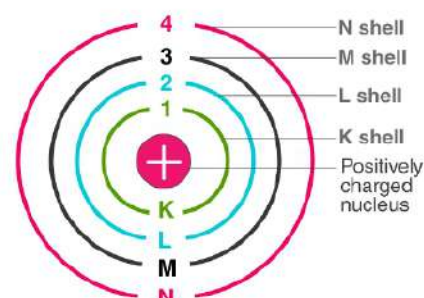
It was derived on the basis of plank's theory.

POSTULATES:

- The electron in the hydrogen atom revolves around the nucleus with definite velocity in fixed, classes circular paths, called 'orbits' or 'shells'.



Photoelectric effect



Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



- These are designed as 1, 2, 3, 4 or K, L, M, N from the side of the nucleus (derived from Kepler's law)
- The e⁻ revolves in the orbit whose angular momentum (mvr) is equal to $nh/2\pi$

$$mvr = nh/2\pi$$
- $\Delta E = E_2 - E_1 = h\nu$
 from higher level to lower orbit ————— energy release
 lower to higher ————— 'E' absorbed
- The centripetal force on the electron is balanced by centrifugal force. Each orbit is associated with definite amount of energy. So these are also called energy levels or energy states.

LIMITATIONS OF BOHR'S MODEL :

It failed to explain the line spectra of atoms or ions having more than one e⁻.

It fails to account the fine spectra details (doublet that is two closely spaced lines) of the hydrogen atom.

- ➔ It fails to explain **Zeman effect** (splitting of spectral lines of an atom into a group of fine lines under the influence of magnetic field is called Zeeman effect).

STARK EFFECT : (Electric field)

Bohr model of the hydrogen atom, not only ignores dual behavior of matter but also contradicts Heisenberg uncertainty principle.

QUANTUM NUMBERS:

Four quantum numbers are requiring for the complete explanation of electron in an atom.

- (1) **Principal quantum number**
- (2) **Azimuthal quantum number**
- (3) **magnetic quantum number**
- (4) **Spin quantum number**

PRINCIPAL QUANTUM NUMBER:

It was proposed by Neil's Bohr the value of $n=1,2,3,4$ or K, L, M, N

- ➔ It indicates the size and energy of the orbit.
- ➔ With increase of 'n' size and energy of the orbital increases.

Total no. of orbitals = n^2

Angular momentum of an e⁻ in an orbit = $nh/2\pi$

The maximum number of electrons in any shell - $2n^2$

Bohr - Bury scheme

Shell	$2n^2$
K	2
L	8
M	18
N	32
O	50



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

AZIMUTHAL QUANTUM NUMBER (l)

:

- Proposed by Somerfield
- The value of (l) depends on "n"
- the values are = 0, 1, 2 (n-1)
- The energies are in the order of $s < p < d < f$
- It indicates the shape of an orbital and angular momentum of electron

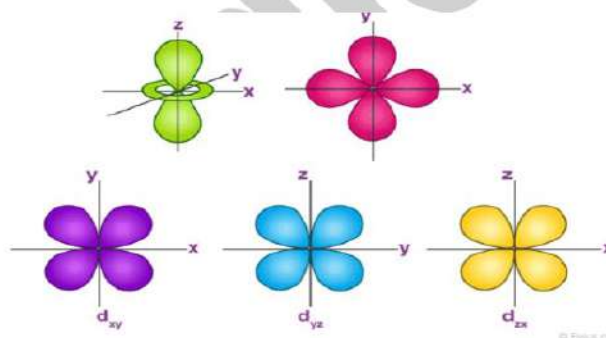
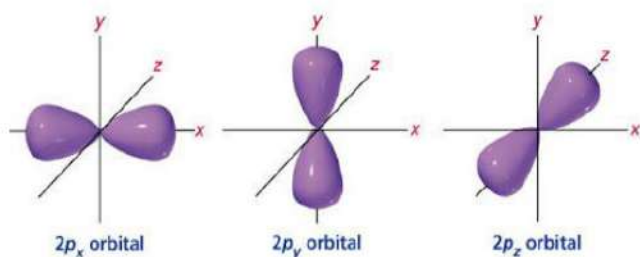
S = Spherical

P = dumbbell

d = double dumbbell

f = complex (or) not define

order of size of different orbitals = $p > sp^3 > sp^2 > sp > s$



NODAL PLANE:

The plane where the probability of finding the electron is zero (= 0) is called Nodal plane.

The number of Nodal planes (or) Angular nodes of an orbital = l

Angular nodes (l)

Eg:	s	0
	p	1
	d	2
	f	3

Number of Radial nodes = n-l-1

Eg:	4s	4-0-1 = 3
	4p	4-1-1 = 2
	4d	4-2-1 = 1



MAGNETIC QUANTUM NUMBER : (Lande)

- The value of 'm' depends on l
- The value of ranges from +l 0 -l
- The total M = 2l +1
- It indicates the orientation of electrons in sub orbits

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics
Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google Play

Download on the
App Store

SPIN QUANTUM NUMBER:

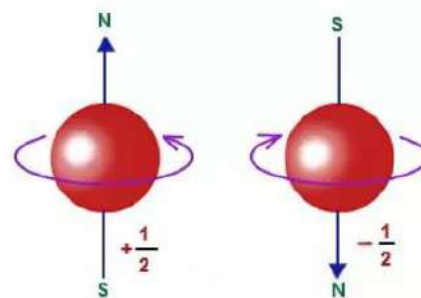
(Goudsmit and whlenbeck)

$$S = +1/2, -1/2$$

The values of spin quantum numbers are independent.

↑ → clockwise direction spin is represented by $+1/2$

↓ → antilock wise - $-1/2$



AUF BAU PRINCIPLE: In the ground state of the atom the orbitals are call filled in order of their ↑ energies.

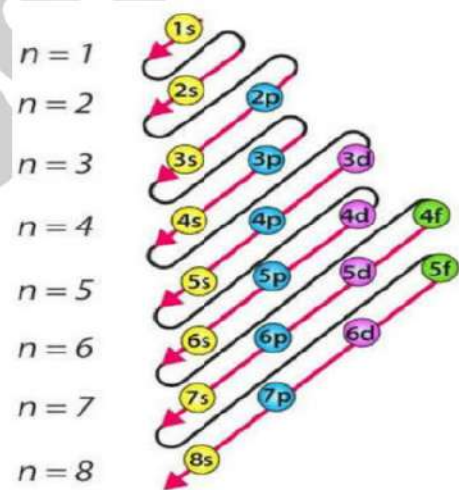
Energy of an orbital = $n+l$

Moeller diagram of order of filling of orbitals.

following element does not follows **Auf- Bau** rule

Example:

Cr	(Ar)	4S03d5
Cu	(Ar)	4S'3d10
Nb	(41)	5So4d5
Mb	(42)	5S' 4d5
Pd	(46)	5S04d10
Ag	(47)	5S'4d10
Gd	(64)	6S25d'4f7
Au	(79)	6S'5d10 4f14



HUND'S RULE:

it states that pairing up of electrons in degenerate orbitals takes place after filling each with one electron of parallel spin.

Follow **Chandan Logics** on



Download **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**

GET IT ON
Google Play

Download on the
App Store



Chandan Logics

Chandan Logics

Aspire to Inspire

Download **Chandan Logics APP**

Contact: 96 76 57 8793

94 94 55 8793

GET IT ON
Google PlayDownload on the
App Store

CARBON ELECTRON CONFIGURATION					
			✓		
1s	2s	2p			
			✓		
1s	2s	2p			
			✗		
1s	2s	2p			
			✗		
1s	2s	2p			

PAULI'S EXCLUSION PRINCIPLE:

it states that no-two electrons in an atom can have the same set of all four quantum numbers.

H = $1s^1$; He = $1s^2$

1st Electrons n=2, l=0, m=0, s = + $\frac{1}{2}$

2nd Electrons n=2, l=0, m=0, s = - $\frac{1}{2}$

Orbital	n	l	m_l	m_s
First electron	1	0	0	+ $\frac{1}{2}$
Second electron	1	0	0	- $\frac{1}{2}$

Follow **Chandan Logics** onDownload **Chandan Logics APP**

Contact: 96 76 57 8793 / 94 94 55 8793

One **STOP** For **ALL Competitive EXAMS**GET IT ON
Google PlayDownload on the
App Store